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SYSTEM AND METHOD FOR DETERMINING AND MAINTAINING DATA QUALITY IN DATA PROCESSING

Abstract

Systems, computer program products, and methods for determining and maintaining data quality in data processing is provided. The method includes determining one or more plugins used in a flow execution. The flow execution performs at least one transformation on one or more data sets. The method also includes determining one or more automatic data quality rules based on at least one of the one or more plugins using in the flow execution. The method further includes determining one or more custom data quality rules based on a custom data quality rule input received from an end-point device. The custom data quality rule includes one or more data statistics to be monitored during the flow execution. The method still further includes causing execution of each of the automatic data quality rules and the one or more custom data quality rules on data in the flow execution.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION [0001] This application is a continuation of and claims the benefit of priority to U.S. patent application Ser. No. 18/244,834, filed Sep. 11, 2023; the contents of which are also hereby incorporated herein.

TECHNOLOGICAL FIELD

[0002] Example embodiments of the present disclosure relate generally to data processing and, more particularly, to determining and maintaining data quality in data processing.

BACKGROUND

[0003] Creating process specific to a given task typically requires coding individual applications. As such, developers are typically required to create flow processes for a network. However, large scale data processing can be hard to monitor to ensure the quality of the data. Through applied effort, ingenuity, and innovation, many of these identified problems have been solved by developing solutions that are included in embodiments of the present disclosure, many examples of which are described in detail herein.

SUMMARY

[0004] The following presents a simplified summary of one or more embodiments of the present disclosure, in order to provide a basic understanding of such embodiments. This summary is not an extensive overview of all contemplated embodiments and is intended to neither identify key or critical elements of all embodiments nor delineate the scope of any or all embodiments. Its sole purpose is to present some concepts of one or more embodiments of the present disclosure in a simplified form as a prelude to the more detailed description that is presented later.

[0005] In an example embodiment, a system for determining and maintaining data quality in data processing is provided. The system includes at least one non-transitory storage device containing instructions and at least one processing device coupled to the at least one non-transitory storage device. The at least one processing device, upon execution of the instructions, is configured to determine one or more plugins used in a flow execution. The flow execution performs at least one transformation on one or more data sets. The at least one processing device, upon execution of the instructions, is also configured to determine one or more automatic data quality rules based on at least one of the one or more plugins using in the flow execution. The at least one processing device, upon execution of the instructions, is further configured to determine one or more custom data quality rules based on a custom data quality rule input received from an end-point device. The custom data quality rule includes one or more data statistics to be monitored during the flow execution. The at least one processing device, upon execution of the instructions, is still further configured to cause execution of each of the automatic data quality rules and the one or more custom data quality rules on data in the flow execution.

[0006] In various embodiments, the at least one processing device, upon execution of the instructions, is configured to cause a rendering of a data quality rule interface configured to receive the custom data quality rule input. In various embodiments, the at least one processing device, upon execution of the instructions, is configured to cause a rendering of results for one or more of

the each of the automatic data quality rules and the one or more custom data quality rules on data in the flow execution to the end-point device.

[0007] In various embodiments, each of the one or more custom data quality rules include a rule category. In various embodiments, the rule category for each of the one or more custom data quality rules is one of accuracy based, completeness based, consistency based, timeliness based, or uniqueness based.

[0008] In various embodiments, each of the one or more automatic data quality rules and the one or more custom data quality rules is one of a column validation, a cross column validation, or a feed level validation. In various embodiments, at least one of the one or more automatic data quality rules or the one or more custom data quality rules is configured to determine whether a data set associated with the flow execution contains a required value.

[0009] In another example embodiment, a computer program product for determining and maintaining data quality in data processing is provided. The computer program product includes at least one non-transitory computer-readable medium having computer-readable program code portions embodied therein. The computer-readable program code portions include one or more executable portions configured to determine one or more plugins used in a flow execution. The flow execution performs at least one transformation on one or more data sets. The computer-readable program code portions include one or more executable portions configured to determine one or more automatic data quality rules based on at least one of the one or more plugins using in the flow execution. The computer-readable program code portions include one or more executable portions configured to determine one or more custom data quality rules based on a custom data quality rule input received from an end-point device. The custom data quality rule includes one or more data statistics to be monitored during the flow execution. The computer-readable program code portions include one or more executable portions configured to cause execution of each of the automatic data quality rules and the one or more custom data quality rules on data in the flow execution.

[0010] In various embodiments, the computer-readable program code portions including one or more executable portions are also configured to cause a rendering of a data quality rule interface configured to receive the custom data quality rule input. In various embodiments, the computer-readable program code portions including one or more executable portions are also configured to cause a rendering of results for one or more of the each of the automatic data quality rules and the one or more custom data quality rules on data in the flow execution to the end-point device.

[0011] In various embodiments, each of the one or more custom data quality rules include a rule category. In various embodiments, the rule category for each of the one or more custom data quality rules is one of accuracy based, completeness based, consistency based, timeliness based, or uniqueness based.

[0012] In various embodiments, each of the one or more automatic data quality rules and the one or more custom data quality rules is one of a column validation, a cross column validation, or a feed level validation. In various embodiments, at least one of the one or more automatic data quality rules or the one or more custom data quality rules is configured to determine whether a data set associated with the flow execution contains a required value.

[0013] In still another example embodiment, a method for determining and maintaining data quality in data processing is provided. The method includes determining one or more plugins used in a flow execution. The flow execution performs at least one transformation on one or more data sets. The method also includes determining one or more automatic data quality rules based on at least one of the one or more plugins using in the flow execution. The method further includes determining one or more custom data quality rules based on a custom data quality rule input received from an end-point device. The custom data quality rule includes one or more data statistics to be monitored during the flow execution. The method still further includes causing execution of each of the automatic data quality rules and the one or more custom data quality rules on data in the

flow execution.

[0014] In various embodiments, the method also includes causing a rendering of a data quality rule interface configured to receive the custom data quality rule input. In various embodiments, the method also includes causing a rendering of results for one or more of the each of the automatic data quality rules and the one or more custom data quality rules on data in the flow execution to the end-point device.

[0015] In various embodiments, each of the one or more custom data quality rules include a rule category. In various embodiments, the rule category for each of the one or more custom data quality rules is one of accuracy based, completeness based, consistency based, timeliness based, or uniqueness based.

[0016] In various embodiments, each of the one or more automatic data quality rules and the one or more custom data quality rules is one of a column validation, a cross column validation, or a feed level validation. In various embodiments, at least one of the one or more automatic data quality rules or the one or more custom data quality rules is configured to determine whether a data set associated with the flow execution contains a required value.

[0017] In an example embodiment, a system for interactive automated code generation and modification for data processing is provided. The system includes at least one non-transitory storage device containing instructions and at least one processing device coupled to the at least one non-transitory storage device. The at least one processing device, upon execution of the instructions, is configured to cause a rendering of a flow designer interface including a flow portion and a plugin portion. The plugin portion includes one or more engageable plugin icons. Each of the one or more engageable plugin icons corresponds to one of one or more plugins. The at least one processing device, upon execution of the instructions, is configured to receive a first plugin input based on a first engagement of a first plugin icon of the one or more engageable plugin icons. The first plugin input is a selection of a first plugin to be included in the flow portion of the flow designer interface. The at least one processing device, upon execution of the instructions, is configured to cause a rendering of a representation of the first plugin corresponding to the first plugin icon on the flow portion of the flow designer interface. The at least one processing device, upon execution of the instructions, is configured to receive a second plugin input based on a second engagement of a second plugin icon of the one or more engagement plugin icons. The second plugin input is a selection of a second plugin to be included in the flow portion of the flow designer interface. The second plugin input includes a connection indication between the first plugin and the second plugin. The at least one processing device, upon execution of the instructions, is configured to cause a rendering of a representation of the second plugin corresponding to the second plugin icon on the flow portion of the flow designer interface. The representation of the first plugin and the representation of the second plugin on the flow portion are connected based on the second plugin input. The at least one processing device, upon execution of the instructions, is configured to generate a flow operation based on the flow portion of the flow designer interface. The flow operation includes each of the first plugin and the second plugin.

[0018] In various embodiments, the at least one processing device, upon execution of the instructions, is configured to cause a rendering of a selected plugin portion of the flow designer interface upon reception of the first plugin input with the selected plugin portion including one or more customizable fields relating to the first plugin.

[0019] In various embodiments, the first plugin icon and the second plugin icon are different plugin icons of the one or more engageable plugin icons. In various embodiments, the first plugin icon and the second plugin icon are a same plugin icon of the one or more engageable plugin icons. In various embodiments, at least one of the first plugin or the second plugin define an input data feed.

[0020] In various embodiments, the at least one processing device, upon execution of the instructions, is configured to receive a third plugin input based on a third engagement of a third plugin icon of the one or more engagement plugin icons with the third plugin input being a

selection of a third plugin to be included in the flow portion of the flow designer interface, and the third plugin input including a connection indication between the third plugin and at least one of the first plugin or the second plugin; and cause a rendering of a representation of the third plugin corresponding to the third plugin icon on the flow portion of the flow designer interface with the representation of the third plugin being connected to at least one of the representation of the first plugin or the representation of the second plugin based on the third plugin input.

[0021] In various embodiments, the at least one processing device, upon execution of the instructions, is configured to cause an execution of the flow operation generated based on the flow portion of the flow designer interface.

[0022] In another example embodiment, a computer program product for interactive automated code generation and modification for data processing is provided. The computer program product includes at least one non-transitory computer-readable medium having computer-readable program code portions embodied therein. The computer-readable program code portions including one or more executable portions configured to cause a rendering of a flow designer interface including a flow portion and a plugin portion. The plugin portion includes one or more engageable plugin icons. Each of the one or more engageable plugin icons corresponds to one of one or more plugins. The computer-readable program code portions including one or more executable portions also configured to receive a first plugin input based on a first engagement of a first plugin icon of the one or more engageable plugin icons. The first plugin input is a selection of a first plugin to be included in the flow portion of the flow designer interface. The computer-readable program code portions including one or more executable portions further configured to cause a rendering of a representation of the first plugin corresponding to the first plugin icon on the flow portion of the flow designer interface. The computer-readable program code portions including one or more executable portions still further configured to receive a second plugin input based on a second engagement of a second plugin icon of the one or more engagement plugin icons. The second plugin input is a selection of a second plugin to be included in the flow portion of the flow designer interface. The second plugin input includes a connection indication between the first plugin and the second plugin. The computer-readable program code portions including one or more executable portions also configured to cause a rendering of a representation of the second plugin corresponding to the second plugin icon on the flow portion of the flow designer interface. The representation of the first plugin and the representation of the second plugin on the flow portion are connected based on the second plugin input. The computer-readable program code portions including one or more executable portions also configured to generate a flow operation based on the flow portion of the flow designer interface. The flow operation includes each of the first plugin and the second plugin.

[0023] In various embodiments, the computer-readable program code portions including one or more executable portions are also configured to cause a rendering of a selected plugin portion of the flow designer interface upon reception of the first plugin input with the selected plugin portion including one or more customizable fields relating to the first plugin.

[0024] In various embodiments, the first plugin icon and the second plugin icon are different plugin icons of the one or more engageable plugin icons. In various embodiments, the first plugin icon and the second plugin icon are a same plugin icon of the one or more engageable plugin icons. In various embodiments, at least one of the first plugin or the second plugin define an input data feed.

[0025] In various embodiments, the computer-readable program code portions including one or more executable portions are also configured to receive a third plugin input based on a third engagement of a third plugin icon of the one or more engagement plugin icons with the third plugin input being a selection of a third plugin to be included in the flow portion of the flow designer interface, and the third plugin input including a connection indication between the third plugin and at least one of the first plugin or the second plugin; and cause a rendering of a representation of the third plugin corresponding to the third plugin icon on the flow portion of the flow designer

interface with the representation of the third plugin being connected to at least one of the representation of the first plugin or the representation of the second plugin based on the third plugin input.

[0026] In various embodiments, the computer-readable program code portions including one or more executable portions are also configured to cause an execution of the flow operation generated based on the flow portion of the flow designer interface.

[0027] In still another example embodiment, a method for interactive automated code generation and modification for data processing is provided. The method includes causing a rendering of a flow designer interface including a flow portion and a plugin portion. The plugin portion includes one or more engageable plugin icons. Each of the one or more engageable plugin icons corresponds to one of one or more plugins. The method also includes receiving a first plugin input based on a first engagement of a first plugin icon of the one or more engageable plugin icons. The first plugin input is a selection of a first plugin to be included in the flow portion of the flow designer interface. The method further includes causing a rendering of a representation of the first plugin corresponding to the first plugin icon on the flow portion of the flow designer interface. The method still further includes receiving a second plugin input based on a second engagement of a second plugin icon of the one or more engagement plugin icons. The second plugin input is a selection of a second plugin to be included in the flow portion of the flow designer interface. The second plugin input includes a connection indication between the first plugin and the second plugin. The method also includes causing a rendering of a representation of the second plugin corresponding to the second plugin icon on the flow portion of the flow designer interface. The representation of the first plugin and the representation of the second plugin on the flow portion are connected based on the second plugin input. The method further includes generating a flow operation based on the flow portion of the flow designer interface. The flow operation includes each of the first plugin and the second plugin.

[0028] In various embodiments, the method includes causing a rendering of a selected plugin portion of the flow designer interface upon reception of the first plugin input with the selected plugin portion including one or more customizable fields relating to the first plugin.

[0029] In various embodiments, the first plugin icon and the second plugin icon are different plugin icons of the one or more engageable plugin icons, or the first plugin icon and the second plugin icon are a same plugin icon of the one or more engageable plugin icons. In various embodiments, at least one of the first plugin or the second plugin define an input data feed.

[0030] In various embodiments, the method includes receiving a third plugin input based on a third engagement of a third plugin icon of the one or more engagement plugin icons with the third plugin input being a selection of a third plugin to be included in the flow portion of the flow designer interface and the third plugin input including a connection indication between the third plugin and at least one of the first plugin or the second plugin; and causing a rendering of a representation of the third plugin corresponding to the third plugin icon on the flow portion of the flow designer interface with the representation of the third plugin being connected to at least one of the representation of the first plugin or the representation of the second plugin based on the third plugin input.

[0031] In various embodiments, the method includes causing an execution of the flow operation generated based on the flow portion of the flow designer interface.

[0032] The features, functions, and advantages that have been discussed may be achieved independently in various embodiments of the present disclosure or may be combined with yet other embodiments, further details of which can be seen with reference to the following description and drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0033] Having described certain example embodiments of the present disclosure in general terms above, reference will now be made to the accompanying drawings. The components illustrated in the figures may or may not be present in certain embodiments described herein. Some embodiments may include fewer (or more) components than those shown in the figures.

[0034] FIGS. **1A-1C** illustrate technical components of an exemplary distributed computing environment for interactive automated code generation and modification for data processing, in accordance with various embodiments of the present disclosure;

[0035] FIG. **2** illustrates an example machine learning (ML) subsystem architecture used to provide interactive automated code generation and modification for data processing, in accordance with various embodiments of the present disclosure;

[0036] FIGS. **3A** and **3B** illustrate a process flow for interactive automated code generation and modification for data processing, in accordance with various embodiments of the present disclosure;

[0037] FIG. **4** illustrates an example feed builder that includes information relating to data feed that may be rendered to a flow designer interface, in accordance with various embodiments of the present disclosure;

[0038] FIG. **5** illustrates another example feed builder that includes information relating to data feed that may be rendered to a flow designer interface, in accordance with various embodiments of the present disclosure;

[0039] FIG. **6** illustrates an example data dictionary that includes information relating to definitions for individual values in a data set that may be rendered to a flow designer interface, in accordance with various embodiments of the present disclosure;

[0040] FIG. **7** illustrates an example flow designer interface that includes a flow and a flow output writer, in accordance with various embodiments of the present disclosure;

[0041] FIG. **8** illustrates an example of flow designer interface **800** including a multi-level grouping functionality, in accordance with various embodiments of the present disclosure;

[0042] FIGS. **9A** and **9B** illustrate an example adjustment plugin, in accordance with various embodiments of the present disclosure;

[0043] FIG. **10** illustrates an example snippet function used to create custom snippet plugins, in accordance with various embodiments of the present disclosure;

[0044] FIG. **11** illustrates a custom snippet plugin being used in a flow, in accordance with various embodiments of the present disclosure;

[0045] FIGS. **12A-12C** illustrate various user interfaces used to generate a data quality rule, in accordance with various embodiments of the present disclosure;

[0046] FIG. **13** illustrates an example data rule dashboard, in accordance with various embodiments of the present disclosure;

[0047] FIGS. **14A-14D** illustrate various user interfaces used to schedule flow executions, in accordance with various embodiments of the present disclosure;

[0048] FIG. **15** illustrates an example execution dashboard, in accordance with various embodiments of the present disclosure;

[0049] FIGS. **16A-16D** illustrate various flow designer interfaces that include data tracing functionality, in accordance with various embodiments of the present disclosure;

[0050] FIG. **17** illustrates a global variable generation interface for defining a global variable, in accordance with various embodiments of the present disclosure;

[0051] FIG. **18** illustrates an example analytics dashboard, in accordance with various embodiments of the present disclosure; and

[0052] FIG. **19** illustrates a process flow for determining and maintaining data quality in data processing.

DETAILED DESCRIPTION

[0053] Embodiments of the present disclosure will now be described more fully hereinafter with reference to the accompanying drawings, in which some, but not all, embodiments of the disclosure are shown. Indeed, the various inventions may be embodied in many different forms and should not be construed as limited to the embodiments set forth herein; rather, these embodiments are provided so that this disclosure will satisfy applicable legal requirements. Where possible, any terms expressed in the singular form herein are meant to also include the plural form and vice versa, unless explicitly stated otherwise. Also, as used herein, the term “a” and/or “an” shall mean “one or more,” even though the phrase “one or more” is also used herein. Furthermore, when it is said herein that something is “based on” something else, it may be based on one or more other things as well. In other words, unless expressly indicated otherwise, as used herein “based on” means “based at least in part on” or “based at least partially on.” Like numbers refer to like elements throughout.

[0054] As used herein, an “entity” may be any institution employing information technology resources and particularly technology infrastructure configured for processing large amounts of data. Typically, these data can be related to the people who work for the organization, its products or services, the customers, or any other aspect of the operations of the organization. As such, the entity may be any institution, group, association, financial institution, establishment, company, union, authority or the like, employing information technology resources for processing large amounts of data.

[0055] As described herein, a “user” may be an individual associated with an entity. As such, in some embodiments, the user may be an individual having past relationships, current relationships or potential future relationships with an entity. In some embodiments, the user may be an employee (e.g., an associate, a project manager, an IT specialist, a manager, an administrator, an internal operations analyst, or the like) of the entity or enterprises affiliated with the entity.

[0056] As used herein, a “user interface” may be a point of human-computer interaction and communication in a device that allows a user to input information, such as commands or data, into a device, or that allows the device to output information to the user. For example, the user interface includes a graphical user interface (GUI) or an interface to input computer-executable instructions that direct a processor to carry out specific functions. The user interface typically employs certain input and output devices such as a display, mouse, keyboard, button, touchpad, touch screen, microphone, speaker, LED, light, joystick, switch, buzzer, bell, and/or other user input/output device for communicating with one or more users.

[0057] As used herein, an “engine” may refer to core elements of an application, or part of an application that serves as a foundation for a larger piece of software and drives the functionality of the software. In some embodiments, an engine may be self-contained, but externally-controllable code that encapsulates powerful logic designed to perform or execute a specific type of function. In one aspect, an engine may be underlying source code that establishes file hierarchy, input and output methods, and how a specific part of an application interacts or communicates with other software and/or hardware. The specific components of an engine may vary based on the needs of the specific application as part of the larger piece of software. In some embodiments, an engine may be configured to retrieve resources created in other applications, which may then be ported into the engine for use during specific operational aspects of the engine. An engine may be configurable to be implemented within any general purpose computing system. In doing so, the engine may be configured to execute source code embedded therein to control specific features of the general purpose computing system to execute specific computing operations, thereby transforming the general purpose system into a specific purpose computing system.

[0058] It should also be understood that “operatively coupled,” as used herein, means that the components may be formed integrally with each other, or may be formed separately and coupled together. Furthermore, “operatively coupled” means that the components may be formed directly to

each other, or to each other with one or more components located between the components that are operatively coupled together. Furthermore, “operatively coupled” may mean that the components are detachable from each other, or that they are permanently coupled together. Furthermore, operatively coupled components may mean that the components retain at least some freedom of movement in one or more directions or may be rotated about an axis (i.e., rotationally coupled, pivotally coupled). Furthermore, “operatively coupled” may mean that components may be electronically connected and/or in fluid communication with one another.

[0059] As used herein, an “interaction” may refer to any communication between one or more users, one or more entities or institutions, one or more devices, nodes, clusters, or systems within the distributed computing environment described herein. For example, an interaction may refer to a transfer of data between devices, an accessing of stored data by one or more nodes of a computing cluster, a transmission of a requested task, or the like.

[0060] As used herein, “determining” may encompass a variety of actions. For example, “determining” may include calculating, computing, processing, deriving, investigating, ascertaining, and/or the like. Furthermore, “determining” may also include receiving (e.g., receiving information), accessing (e.g., accessing data in a memory), and/or the like. Also, “determining” may include resolving, selecting, choosing, calculating, establishing, and/or the like. Determining may also include ascertaining that a parameter matches a predetermined criterion, including that a threshold has been met, passed, exceeded, and so on.

[0061] Automating processing flows typically requires hard coding individual processes. As such, any changes to the processing may cause the other processes in the flow to be changed. However, such operations are time consuming and difficult to achieve. Additionally, non-technical users are not able to build and/or automate processes.

[0062] Various embodiments of the present disclosure provide interactive automated code generation and modification for data processing. As such, various embodiments provide a web-based, visual development platform that does not require individual coding of plugins. The modularized platform encompasses data ingestion, data quality checks, flow creation, flow execution scheduling, calculation integration, and/or the like. To do this, the system provides a flow designer interface with a plugin portion and a flow portion of the interface. The plugin portion includes each of the plugins that are available to be added to the flow. Selected plugins are then rendered in the flow portion. Multiple plugins can be represented in the flow portion and connected to one another to create a flow. The operations necessary to execute the flow are generated based on the plugin representations rendered in the flow portion of the flow designer interface. Various embodiments discussed herein allow for the flow designer interface to be customized and/or updated for different use cases.

[0063] Determining data feeds to use for a flow can be difficult and time consuming for users. As the complexity of data processing increases, the number of different data sources increases. Additionally, the data sources may update at different times, causing a system to either have to continuously monitor for data changes or potentially be behind real-time (e.g., the data may be out of date).

[0064] Various embodiments of the present disclosure are used for determining data feed sources for interactive automated code generation and modification. The system may provide an interface for selecting data sources to be included during data processing. The data sources may each be determined by the system and listed for a user to select the sources to be used during processing. The system may provide recommended data feeds based on the data feeds available. The user can make changes to existing data feeds and/or add new data feeds not previously identified. Additionally, the system may monitor for different key event indications to occur in order to cause data import to be actuated. For example, the system may communicate with one or more data sources to determine when the data source has been updated. As such, the data importing does not have to be continuous, but does not lag real-time. A network user may designate key event

indications, allowing the system to be customizable.

[0065] Large scale data processing, especially from different sources can result in undesirable results for various different reasons. As such, processed data needs to be monitored to ensure the desired output is being produced. Example issues with data processing may include data formatting (e.g., different data sources may have different data formatting), unexpected data size, incomplete data sets, data scaling (e.g., value units), and/or the like. Typically, processed data has to be monitored to ensure the accuracy of outputs. However, in large scale processes, it can be difficult to determine specific causes of undesirable results without individually investigating each individual step of the process. As such, processing is either delayed to ensure data quality or executed with reduced data quality standards.

[0066] Various embodiments of the present disclosure allow for determining and maintaining data quality in data processing. To do this, the system provides the ability to implement data quality rules for data processing that can ensure that the data processing is operating correctly. The system may determine automatic data quality rules that are assigned for a flow. The system may provide a user interface in which a network user can create custom data quality rules. The network user may view and/or edit any of the data quality rules assigned for given flow via an end-point device. The system may also provide a data quality dashboard that provides information relating to a data process (e.g., time to execute a data flow, any potential errors, etc.).

[0067] FIGS. 1A-1C illustrate technical components of an exemplary distributed computing environment for interactive automated code generation and modification for data processing, in accordance with various embodiments of the disclosure. As shown in FIG. 1A, the distributed computing environment **100** contemplated herein may include a system **130** (e.g., a flow designer generation device), an end-point device(s) **140**, and one or more networks **110** over which the system **130** and end-point device(s) **140** communicate therebetween. FIG. 1A illustrates only one example of an embodiment of the distributed computing environment **100**, and it will be appreciated that in other embodiments one or more of the systems, devices, and/or servers may be combined into a single system, device, or server, or be made up of multiple systems, devices, or servers. Also, the distributed computing environment **100** may include multiple systems, same or similar to system **130**, with each system providing portions of the necessary operations (e.g., as a server bank, a group of blade servers, or a multi-processor system).

[0068] In some embodiments, the system **130** and the end-point device(s) **140** may have a client-server relationship in which the end-point device(s) **140** are remote devices that request and receive service from a centralized server, i.e., the system **130**. In some other embodiments, the system **130** and the end-point device(s) **140** may have a peer-to-peer relationship in which the system **130** and the end-point device(s) **140** are considered equal and all have the same abilities to use the resources available on the network(s) **110**. Instead of having a central server (e.g., system **130**) which would act as the shared drive, each device that is connect to the network(s) **110** would act as the server for the files stored on it.

[0069] The system **130** may represent various forms of servers, such as web servers, database servers, file server, or the like, various forms of digital computing devices, such as laptops, desktops, video recorders, audio/video players, radios, workstations, or the like, or any other auxiliary network devices, such as wearable devices, Internet-of-things devices, electronic kiosk devices, mainframes, or the like, or any combination of the aforementioned.

[0070] The end-point device(s) **140** may represent various forms of electronic devices, including user input devices such as personal digital assistants, cellular telephones, smartphones, laptops, desktops, and/or the like, merchant input devices such as point-of-sale (POS) devices, electronic payment kiosks, and/or the like, electronic telecommunications device (e.g., automated teller machine (ATM)), and/or edge devices such as routers, routing switches, integrated access devices (IAD), and/or the like.

[0071] The network(s) **110** may be a distributed network that is spread over different networks.

This provides a single data communication network, which can be managed jointly or separately by each network. Besides shared communication within the network, the distributed network often also supports distributed processing. The network(s) **110** may be a form of digital communication network such as a telecommunication network, a local area network (“LAN”), a wide area network (“WAN”), a global area network (“GAN”), the Internet, satellite network, cellular network, and/or any combination of the foregoing. The network(s) **110** may be secure and/or unsecure and may also include wireless and/or wired and/or optical interconnection technology.

[0072] It is to be understood that the structure of the distributed computing environment and its components, connections and relationships, and their functions, are meant to be exemplary only, and are not meant to limit implementations of the disclosure described and/or claimed in this document. In one example, the distributed computing environment **100** may include more, fewer, or different components. In another example, some or all of the portions of the distributed computing environment **100** may be combined into a single portion or all of the portions of the system **130** may be separated into two or more distinct portions.

[0073] FIG. **1B** illustrates an exemplary component-level structure of the system **130**, in accordance with an embodiment of the disclosure. As shown in FIG. **1B**, the system **130** may include a processor **102**, memory **104**, input/output (I/O) device **116**, and a storage device **106**. The system **130** may also include a high-speed interface **108** connecting to the memory **104**, and a low-speed interface **112** (shown as “LS Interface”) connecting to low-speed expansion port **114** (shown as “LS Port”) and storage device **110**. Each of the components **102**, **104**, **106**, **108**, **110**, and **112** may be operatively coupled to one another using various buses and may be mounted on a common motherboard or in other manners as appropriate. As described herein, the processor **102** may include a number of subsystems to execute the portions of processes described herein. Each subsystem may be a self-contained component of a larger system (e.g., system **130**) and capable of being configured to execute specialized processes as part of the larger system.

[0074] The processor **102** can process instructions, such as instructions of an application that may perform the functions disclosed herein. These instructions may be stored in the memory **104** (e.g., non-transitory storage device) or on the storage device **106**, for execution within the system **130** using any subsystems described herein. It is to be understood that the system **130** may use, as appropriate, multiple processors, along with multiple memories, and/or I/O devices, to execute the processes described herein.

[0075] The memory **104** stores information within the system **130**. In one implementation, the memory **104** is a volatile memory unit or units, such as volatile random access memory (RAM) having a cache area for the temporary storage of information, such as a command, a current operating state of the distributed computing environment **100**, an intended operating state of the distributed computing environment **100**, instructions related to various methods and/or functionalities described herein, and/or the like. In another implementation, the memory **104** is a non-volatile memory unit or units. The memory **104** may also be another form of computer-readable medium, such as a magnetic or optical disk, which may be embedded and/or may be removable. The non-volatile memory may additionally or alternatively include an EEPROM, flash memory, and/or the like for storage of information such as instructions and/or data that may be read during execution of computer instructions. The memory **104** may store, recall, receive, transmit, and/or access various files and/or information used by the system **130** during operation.

[0076] The storage device **106** is capable of providing mass storage for the system **130**. In one aspect, the storage device **106** may be or contain a computer-readable medium, such as a floppy disk device, a hard disk device, an optical disk device, or a tape device, a flash memory or other similar solid state memory device, or an array of devices, including devices in a storage area network or other configurations. A computer program product can be tangibly embodied in an information carrier. The computer program product may also contain instructions that, when executed, perform one or more methods, such as those described above. The information carrier

may be a non-transitory computer-or machine-readable storage medium, such as the memory **104**, the storage device **106**, or memory on processor **102**.

[0077] The high-speed interface **108** manages bandwidth-intensive operations for the system **130**, while the low-speed interface **112** manages lower bandwidth-intensive operations. Such allocation of functions is exemplary only. In some embodiments, the high-speed interface **108** (shown as “HS Interface”) is coupled to memory **104**, input/output (I/O) device **116** (e.g., through a graphics processor or accelerator), and to high-speed expansion ports **111** (shown as “HS Port”), which may accept various expansion cards (not shown). In such an implementation, low-speed interface **112** is coupled to storage device **106** and low-speed expansion port **114**. The low-speed expansion port **114**, which may include various communication ports (e.g., USB, Bluetooth, Ethernet, wireless Ethernet), may be coupled to one or more input/output devices, such as a keyboard, a pointing device, a scanner, or a networking device such as a switch or router, e.g., through a network adapter.

[0078] The system **130** may be implemented in a number of different forms. For example, it may be implemented as a standard server, or multiple times in a group of such servers. Additionally, the system **130** may also be implemented as part of a rack server system or a personal computer such as a laptop computer. Alternatively, components from system **130** may be combined with one or more other same or similar systems and an entire system **130** may be made up of multiple computing devices communicating with each other.

[0079] FIG. **1C** illustrates an exemplary component-level structure of the end-point device(s) **140**, in accordance with an embodiment of the disclosure. As shown in FIG. **1C**, the end-point device(s) **140** includes a processor **152**, memory **154**, an input/output device such as a display **156**, a communication interfaces **158**, and a transceiver **160**, among other components. The end-point device(s) **140** may also be provided with a storage device, such as a micro-drive or other device, to provide additional storage. Each of the components **152**, **154**, **158**, and **160**, are interconnected using various buses, and several of the components may be mounted on a common motherboard or in other manners as appropriate.

[0080] The processor **152** is configured to execute instructions within the end-point device(s) **140**, including instructions stored in the memory **154**, which in one embodiment includes the instructions of an application that may perform the functions disclosed herein, including certain logic, data processing, and data storing functions. The processor may be implemented as a chipset of chips that include separate and multiple analog and digital processors. The processor may be configured to provide, for example, for coordination of the other components of the end-point device(s) **140**, such as control of user interfaces, applications run by end-point device(s) **140**, and wireless communication by end-point device(s) **140**.

[0081] The processor **152** may be configured to communicate with the user through control interface **164** and display interface **166** coupled to a display **156**. The display **156** may be, for example, a TFT LCD (Thin-Film-Transistor Liquid Crystal Display) or an OLED (Organic Light Emitting Diode) display, or other appropriate display technology. The display **156** may comprise appropriate circuitry and configured for driving the display **156** to present graphical and other information to a user. The control interface **164** may receive commands from a user and convert them for submission to the processor **152**. In addition, an external interface **168** may be provided in communication with processor **152**, so as to enable near area communication of end-point device(s) **140** with other devices. External interface **168** may provide, for example, for wired communication in some implementations, or for wireless communication in other implementations, and multiple interfaces may also be used.

[0082] The memory **154** stores information within the end-point device(s) **140**. The memory **154** can be implemented as one or more of a computer-readable medium or media, a volatile memory unit or units, or a non-volatile memory unit or units. Expansion memory may also be provided and connected to end-point device(s) **140** through an expansion interface (not shown), which may

include, for example, a SIMM (Single In Line Memory Module) card interface. Such expansion memory may provide extra storage space for end-point device(s) **140** or may also store applications or other information therein. In some embodiments, expansion memory may include instructions to carry out or supplement the processes described above and may include secure information also. For example, expansion memory may be provided as a security module for end-point device(s) **140** and may be programmed with instructions that permit secure use of end-point device(s) **140**. In addition, secure applications may be provided via the SIMM cards, along with additional information, such as placing identifying information on the SIMM card in a non-hackable manner. [0083] The memory **154** may include, for example, flash memory and/or NVRAM memory. In one aspect, a computer program product is tangibly embodied in an information carrier. The computer program product contains instructions that, when executed, perform one or more methods, such as those described herein. The information carrier is a computer-or machine-readable medium, such as the memory **154**, expansion memory, memory on processor **152**, or a propagated signal that may be received, for example, over transceiver **160** or external interface **168**.

[0084] In some embodiments, the user may use the end-point device(s) **140** to transmit and/or receive information or commands to and from the system **130** via the network(s) **110**. Any communication between the system **130** and the end-point device(s) **140** may be subject to an authentication protocol allowing the system **130** to maintain security by permitting only authenticated users (or processes) to access the protected resources of the system **130**, which may include servers, databases, applications, and/or any of the components described herein. To this end, the system **130** may trigger an authentication subsystem that may require the user (or process) to provide authentication credentials to determine whether the user (or process) is eligible to access the protected resources. Once the authentication credentials are validated and the user (or process) is authenticated, the authentication subsystem may provide the user (or process) with permissioned access to the protected resources. Similarly, the end-point device(s) **140** may provide the system **130** (or other client devices) permissioned access to the protected resources of the end-point device(s) **140**, which may include a GPS device, an image capturing component (e.g., camera), a microphone, and/or a speaker.

[0085] The end-point device(s) **140** may communicate with the system **130** through at least one of communication interfaces **158**, which may include digital signal processing circuitry where necessary. Communication interfaces **158** may provide for communications under various modes or protocols, such as the Internet Protocol (IP) suite (commonly known as TCP/IP). Protocols in the IP suite define end-to-end data handling methods for everything from packetizing, addressing, and routing, to receiving. Broken down into layers, the IP suite includes the link layer, containing communication methods for data that remains within a single network segment (link); the Internet layer, providing internetworking between independent networks; the transport layer, handling host-to-host communication; and the application layer, providing process-to-process data exchange for applications. Each layer contains a stack of protocols used for communications. In addition, the communication interfaces **158** may provide for communications under various telecommunications standards (2G, 3G, 4G, 5G, and/or the like) using their respective layered protocol stacks. These communications may occur through a transceiver **160**, such as radio-frequency transceiver. In addition, short-range communication may occur, such as using a Bluetooth, Wi-Fi, or other such transceiver (not shown). In addition, GPS (Global Positioning System) receiver module **170** may provide additional navigation-and location-related wireless data to end-point device(s) **140**, which may be used as appropriate by applications running thereon, and in some embodiments, one or more applications operating on the system **130**. The end-point device(s) **140** may include a communication interface that is configured to operate with a satellite network.

[0086] In various embodiments, the end-point device(s) **140** may have multiple communication interfaces that are configured to operate using the various communication methods discussed herein. For example, an end-point device **140** may have a cellular network communication

interface (e.g., a communication interface that provides for communication under various telecommunications standards) and a satellite network communication interface (e.g., a communication interface that provides for communication via a satellite network). Various other communication interfaces may also be provided by the end-point device (e.g., an end-point device may be capable of communicating via a cellular network, a satellite network, and/or a wi-fi connection). Various communication interfaces may share components with other communication interfaces in the given end-point device.

[0087] The end-point device(s) **140** may also communicate audibly using audio codec **162**, which may receive spoken information from a user and convert it to usable digital information. Audio codec **162** may likewise generate audible sound for a user, such as through a speaker, e.g., in a handset of end-point device(s) **140**. Such sound may include sound from voice telephone calls, may include recorded sound (e.g., voice messages, music files, etc.) and may also include sound generated by one or more applications operating on the end-point device(s) **140**, and in some embodiments, one or more applications operating on the system **130**.

[0088] Various implementations of the distributed computing environment **100**, including the system **130** and end-point device(s) **140**, and techniques described here can be realized in digital electronic circuitry, integrated circuitry, specially designed ASICs (application specific integrated circuits), computer hardware, firmware, software, and/or combinations thereof.

[0089] FIG. **2** illustrates an example machine learning (ML) architecture **200**, in accordance with an embodiment of the present disclosure. The ML subsystem architecture may be part of the components of the environment **100** (e.g., system **130**). The ML subsystem architecture is used to generate various different components of the flow designer interface.

[0090] The machine learning subsystem **200** may include a data acquisition engine **202**, data ingestion engine **210**, data pre-processing engine **216**, ML model tuning engine **222**, and inference engine **236**.

[0091] The data acquisition engine **202** may identify various internal and/or external data sources to generate, test, and/or integrate new features for training the machine learning model **224**. These internal and/or external data sources **204**, **206**, and **208** may be initial locations where the data originates or where physical information is first digitized. The data acquisition engine **202** may identify the location of the data and describe connection characteristics for access and retrieval of data. In some embodiments, data is transported from each data source **204**, **206**, or **208** using any applicable network protocols, such as the File Transfer Protocol (FTP), Hyper-Text Transfer Protocol (HTTP), or any of the myriad Application Programming Interfaces (APIs) provided by websites, networked applications, and other services. In some embodiments, these data sources **204**, **206**, and **208** may include Enterprise Resource Planning (ERP) databases that host data related to day-to-day business activities such as accounting, procurement, project management, exposure management, supply chain operations, and/or the like, mainframe that is often the entity's central data processing center, edge devices that may be any piece of hardware, such as sensors, actuators, gadgets, appliances, or machines, that are programmed for certain applications and can transmit data over the internet or other networks, and/or the like. The data acquired by the data acquisition engine **202** from these data sources **204**, **206**, and **208** may then be transported to the data ingestion engine **210** for further processing.

[0092] Depending on the nature of the data imported from the data acquisition engine **202**, the data ingestion engine **210** may move the data to a destination for storage or further analysis. Typically, the data imported from the data acquisition engine **202** may be in varying formats as they come from different sources, including RDBMS, other types of databases, S3 buckets, CSVs, or from streams. Since the data comes from different places, it needs to be cleansed and transformed so that it can be analyzed together with data from other sources. At the data ingestion engine **210**, the data may be ingested in real-time, using the stream processing engine **212**, in batches using the batch data warehouse **214**, or a combination of both. The stream processing engine **212** may be used to

process continuous data stream (e.g., data from edge devices), i.e., computing on data directly as it is received, and filter the incoming data to retain specific portions that are deemed useful by aggregating, analyzing, transforming, and ingesting the data. On the other hand, the batch data warehouse **214** collects and transfers data in batches according to scheduled intervals, trigger events, or any other logical ordering.

[0093] In machine learning, the quality of data and the useful information that can be derived therefrom directly affects the ability of the machine learning model **224** to learn. The data pre-processing engine **216** may implement advanced integration and processing steps needed to prepare the data for machine learning execution. This may include modules to perform any upfront, data transformation to consolidate the data into alternate forms by changing the value, structure, or format of the data using generalization, normalization, attribute selection, and aggregation, data cleaning by filling missing values, smoothing the noisy data, resolving the inconsistency, and removing outliers, and/or any other encoding steps as needed.

[0094] In addition to improving the quality of the data, the data pre-processing engine **216** may implement feature extraction and/or selection techniques to generate training data **218**. Feature extraction and/or selection is a process of dimensionality reduction by which an initial set of data is reduced to more manageable groups for processing. A characteristic of these large data sets is a large number of variables that require a lot of computing resources to process. Feature extraction and/or selection may be used to select and/or combine variables into features, effectively reducing the amount of data that must be processed, while still accurately and completely describing the original data set. Depending on the type of machine learning algorithm being used, this training data **218** may require further enrichment. For example, in supervised learning, the training data is enriched using one or more meaningful and informative labels to provide context so a machine learning model can learn from it. For example, labels might indicate whether a photo contains a bird or car, which words were uttered in an audio recording, or if an x-ray contains a tumor. Data labeling is required for a variety of use cases including computer vision, natural language processing, and speech recognition. In contrast, unsupervised learning uses unlabeled data to find patterns in the data, such as inferences or clustering of data points.

[0095] The ML model tuning engine **222** may be used to train a machine learning model **224** using the training data **218** to make predictions or decisions without explicitly being programmed to do so. The machine learning model **224** represents what was learned by the selected machine learning algorithm **220** and represents the rules, numbers, and any other algorithm-specific data structures required for classification. Selecting the right machine learning algorithm may depend on a number of different factors, such as the problem statement and the kind of output needed, type and size of the data, the available computational time, number of features and observations in the data, and/or the like. Machine learning algorithms may refer to programs (math and logic) that are configured to self-adjust and perform better as they are exposed to more data. To this extent, machine learning algorithms are capable of adjusting their own parameters, given feedback on previous performance in making prediction about a dataset.

[0096] The machine learning algorithms contemplated, described, and/or used herein include supervised learning (e.g., using logistic regression, using back propagation neural networks, using random forests, decision trees, etc.), unsupervised learning (e.g., using an Apriori algorithm, using K-means clustering), semi-supervised learning, reinforcement learning (e.g., using a Q-learning algorithm, using temporal difference learning), and/or any other suitable machine learning model type. Each of these types of machine learning algorithms can implement any of one or more of a regression algorithm (e.g., ordinary least squares, logistic regression, stepwise regression, multivariate adaptive regression splines, locally estimated scatterplot smoothing, etc.), an instance-based method (e.g., k-nearest neighbor, learning vector quantization, self-organizing map, etc.), a regularization method (e.g., ridge regression, least absolute shrinkage and selection operator, elastic net, etc.), a decision tree learning method (e.g., classification and regression tree, iterative

dichotomiser 3, C4.5, chi-squared automatic interaction detection, decision stump, random forest, multivariate adaptive regression splines, gradient boosting machines, etc.), a Bayesian method (e.g., naïve Bayes, averaged one-dependence estimators, Bayesian belief network, etc.), a kernel method (e.g., a support vector machine, a radial basis function, etc.), a clustering method (e.g., k-means clustering, expectation maximization, etc.), an associated rule learning algorithm (e.g., an Apriori algorithm, an Eclat algorithm, etc.), an artificial neural network model (e.g., a Perceptron method, a back-propagation method, a Hopfield network method, a self-organizing map method, a learning vector quantization method, etc.), a deep learning algorithm (e.g., a restricted Boltzmann machine, a deep belief network method, a convolution network method, a stacked auto-encoder method, etc.), a dimensionality reduction method (e.g., principal component analysis, partial least squares regression, Sammon mapping, multidimensional scaling, projection pursuit, etc.), an ensemble method (e.g., boosting, bootstrapped aggregation, AdaBoost, stacked generalization, gradient boosting machine method, random forest method, etc.), and/or the like.

[0097] To tune the machine learning model, the ML model tuning engine **222** may repeatedly execute cycles of experimentation **226**, testing **228**, and tuning **230** to optimize the performance of the machine learning algorithm **220** and refine the results in preparation for deployment of those results for consumption or decision making. To this end, the ML model tuning engine **222** may dynamically vary hyperparameters each iteration (e.g., number of trees in a tree-based algorithm or the value of alpha in a linear algorithm), run the algorithm on the data again, then compare its performance on a validation set to determine which set of hyperparameters results in the most accurate model. The accuracy of the model is the measurement used to determine which set of hyperparameters is best at identifying relationships and patterns between variables in a dataset based on the input, or training data **218**. A fully trained machine learning model **232** is one whose hyperparameters are tuned and model accuracy maximized.

[0098] The trained machine learning model **232**, similar to any other software application output, can be persisted to storage, file, memory, or application, or looped back into the processing component to be reprocessed. More often, the trained machine learning model **232** is deployed into an existing production environment to make practical business decisions based on live data **234**. To this end, the machine learning subsystem **200** uses the inference engine **236** to make such decisions. The type of decision-making may depend upon the type of machine learning algorithm used. For example, machine learning models trained using supervised learning algorithms may be used to structure computations in terms of categorized outputs (e.g., C_1, C_2 . . . C_n **238**) or observations based on defined classifications, represent possible solutions to a decision based on certain conditions, model complex relationships between inputs and outputs to find patterns in data or capture a statistical structure among variables with unknown relationships, and/or the like. On the other hand, machine learning models trained using unsupervised learning algorithms may be used to group (e.g., C_1, C_2 . . . C_n **238**) live data **234** based on how similar they are to one another to solve exploratory challenges where little is known about the data, provide a description or label (e.g., C_1, C_2 . . . C_n **238**) to live data **234**, such as in classification, and/or the like. These categorized outputs, groups (clusters), or labels are then presented to the user input system **130**. In still other cases, machine learning models that perform regression techniques may use live data **234** to predict or forecast continuous outcomes.

[0099] It will be understood that the embodiment of the machine learning subsystem **200** illustrated in FIG. 2 is example and that other embodiments may vary. As another example, in some embodiments, the machine learning subsystem **200** may include more, fewer, or different components.

[0100] FIGS. 3A and 3B include a flow chart **300** that illustrates an example method of providing interactive automated code generation and modification for data processing. The method may be carried out by various components of the distributed computing environment **100** discussed herein (e.g., the system **130**, one or more end-point devices **140**, etc.). An example system may include at

least one non-transitory storage device and at least one processing device coupled to the at least one non-transitory storage device. In such an embodiment, the at least one processing device is configured to carry out the method discussed herein. A method of various embodiments may include any combination or subset of the features discussed herein.

[0101] Referring now to Block **302** of FIG. 3A, the method includes causing a rendering of a flow designer interface including a flow portion and a plugin portion. An example flow portion is the flow portion **725** shown in FIG. 7. The flow portion may be used to create a flow operation using block (e.g., Blocks **701**, **702**, **703**, **704**, **705** of FIG. 7) that are connected to one another to define a flow. Each of the blocks represent a plugin (e.g., a plugin selected from the plugin portion **705**).

[0102] An example plugin portion is the plugin portion **705** shown in FIG. 7. The plugin portion includes one or more engagement plugin icons. Each of the engagement plugin icon(s) correspond to a plugin. Example plugin icons shown in FIG. 7 include file loader, SQL loader, data aggregator, feed loader, calc service, data cleanse, enricher, join datasets, among others in the plugin portion **705**. Each of the plugins may designate data feeds (e.g., file loader, SQL loader, data aggregator, feed loader, etc.) and/or perform transformations and/or calculations relating to the data sets (e.g., calc service, data cleanse, enricher, join datasets, etc.). In various embodiments, the different plugin types (e.g., input data feed, transformations, etc.) may be designated using different icon coloring (e.g., the input data feeds may be green, and the data transformation may be purple).

[0103] Referring now to Block **304** of FIG. 3A, the method includes receiving a first plugin input based on a first engagement of a first plugin icon of the one or more engageable plugin icons. The first plugin input is a selection of a first plugin to be included in the flow portion of the flow designer interface. In various embodiments, the first plugin input may be an engagement with one of the plugin icons of the plugin portion. The engagement may be a click or drag (e.g., a user may drag the plugin icon to the flow portion of the flow designer interface).

[0104] The first plugin input may include the selection of the plugin (e.g., which plugin icon from the plugin portion to be selected to be used in the flow portion). Additionally, the first plugin input may include information relating to the location of the first plugin in the flow (e.g., the plugin icon may be drag to the location on the flow portion in which the first plugin is to be rendered).

[0105] In various embodiments, the first plugin input and/or inputs received after the first plugin input may include information relating to the first plugin, such as any data feeds, any variables used, and/or the like. In various embodiments, the first plugin may have a predefined value for each of the inputs that are customizable (e.g., via an input from a network user).

[0106] Referring now to Block **306** of FIG. 3A, the method includes causing a rendering of a representation of the first plugin corresponding to the first plugin icon on the flow portion of the flow designer interface. Based on the first plugin input, the system causes a representation of the first plugin to be displayed on the flow portion. For example, a first plugin may be represented as a block as shown in FIG. 7 (e.g., Blocks **701**, **702**, **703**, etc.). In various embodiments, the representation of the first plugin may be engageable in the flow portion (e.g., able to connect with other plugin representations and/or access the selected plugin portion of the flow designer interface).

[0107] Referring now to optional Block **308** of FIG. 3A, the method includes causing a rendering of a selected plugin portion of the flow designer interface upon reception of the first plugin input. An example selected plugin portion **730** is shown in FIG. 7. In various embodiments, the selected plugin portion **730** may include information relating to a selected plugin. The selected plugin portion may include information relating to any of the plugins shown on the flow portion of the flow designer interface. In various embodiments, upon selecting a plugin to be rendered on the flow portion of the flow designer interface (e.g., first plugin, second plugin, third plugin, etc.), the selected plugin portion may display information relating to the given plugin. Another example selected plugin portion is the adjustment portion **920** in which information relating to the adjustment plugin **910** is shown.

[0108] Additionally or alternatively, the selected plugin portion may include information relating to the plugin corresponding to a representation in the flow portion that is engaged. For example, in an example embodiment in which the representation of the first plugin (and/or other plugins) is engageable in the flow portion, upon engagement, information relating to the first plugin (or other selected plugin) may be displayed via a selected plugin portion of the flow designer.

[0109] In various embodiments, the selected plugin portion may include one or more customizable fields for the given plugin (e.g., first plugin, second plugin, etc.). For example, a network user may be able to customize certain features of a plugin, such as the variables, data feeds, and/or the like. Additionally, the selected plugin portion may include other information relating to the selected plugin, such as the input data and/or the output data for the plugin.

[0110] Referring now to Block **310** of FIG. **3A**, the method includes receiving a second plugin input based on a second engagement of a second plugin icon of the one or more engagement plugin icons. The second engagement of the second plugin icon may be the same or similar to the first engagement of the first plugin icon. As such, the second plugin input is a selection of a second plugin to be included in the flow portion of the flow designer interface.

[0111] In various embodiments, the first plugin icon and the second plugin icon may be different plugin icons of the one or more engageable plugin icons. Alternatively, the first plugin icon and the second plugin icon may be a same plugin icon of the one or more engageable plugin icons. In various embodiments, each of the first plugin and the second plugin may be various types of plugins (e.g., transformation, data source, etc.). For example, at least one of the first plugin or the second plugin may define an input data feed. In various embodiments, the flow portion may include more than a first plugin and a second plugin. As such, the first plugin and the second plugin may not necessarily be the first and second plugins included in a flow.

[0112] The second plugin input may also include a connection indication between the first plugin and the second plugin. The connection indication may be received at the same time as the second engagement of the second plugin icon or separate from the second engagement of the second plugin icon (e.g., the system may prompt the network user to provide the connection indication that indicates the connection between the first plugin and second plugin). The connection indication may include information relating to how the selected plugin should be connected to one or more other plugins. For example, a line may be drawn on the flow portion of the flow designer interface between one or more plugins on the flow portion to indicate the flow of data during execution.

[0113] In various embodiments, the AI/ML model(s) may provide recommendations on how each plugin interconnects. For example, the AI/ML model(s) may recommend connections between plugins based on previous flow designs (e.g., in previous flow designs, users may often connect certain plugins together). Additionally or alternatively, the AI/ML model(s) may recommend connections between the plugins based on the type of plugins. For example, certain transformation type plugins may often be connected to a data source type plugin.

[0114] Referring now to Block **312** of FIG. **3A**, the method includes causing a rendering of a representation of the second plugin corresponding to the second plugin icon on the flow portion of the flow designer interface. The representation of the second plugin may be represented in the same or similar way that the first plugin is represented in the flow portion of the flow designer interface.

[0115] In various embodiments, the flow portion of the flow designer interface may connections between the renderings of the various plugins, such that the flow of data is shown between plugins. For example, the representation of the first plugin and the representation of the second plugin on the flow portion may be connected based on the second plugin input. The connection may be represented as an arrow with the direction of the arrow indicating the direction in which data moves through the plugins during the flow execution. For example, in an instance the arrow is pointing from the representation of the first plugin in the direction of the representation of the second plugin, the data moves from the first plugin to the second plugin during operation. In various embodiments, a double ended arrow may indicate that some data in the flow is moving in both

directions (e.g., the first plugin may process data, then the second plugin may process the data processed by the first plugin, and then the first plugin may again process the data processed by the second plugin).

[0116] Referring now to optional Block **314** of FIG. **3B**, the method includes receiving a third plugin input based on a third engagement of a third plugin icon of the one or more engagement plugin icons. Just as the first plugin input and the second input selection, the third plugin input is a selection of a plugin (e.g., a third plugin) to be included in the flow portion of the flow designer interface. The third plugin may be the same or different than either of the first plugin or the second plugin.

[0117] The third plugin input may also include a connection indication between the third plugin and other plugins represented in the flow portion (e.g., at least one of the first plugin or the second plugin). The connection indication for the third plugin may be received at the same time as the third engagement of the third plugin icon or separate from the third engagement of the third plugin icon (e.g., the system may prompt the network user to provide the connection indication that indicates the connection other plugins). The connection indication may include information relating to how the selected plugin should be connected to one or more other plugins. For example, a line may be drawn on the flow portion of the flow designer interface between one or more plugins on the flow portion to indicate the flow of data during execution.

[0118] In various embodiments, additional engagements may indicate additional plugins to be represented in the flow portion. For example, a fourth engagement of a fourth plugin icon may indicate a fourth plugin to be included in the flow portion, a fifth engagement of a fifth plugin icon may indicate a fifth plugin to be included in the flow portion, and/or the like.

[0119] Referring now to optional Block **316** of FIG. **3B**, the method includes causing a rendering of a representation of the third plugin corresponding to the third plugin icon on the flow portion of the flow designer interface. The third plugin may be represented in the flow portion of the flow designer interface, just as the first plugin and/or the second plugin are represented. The third plugin may be the same or a different plugin than the first plugin and/or the second plugin. As such, two or more of the first plugin, second plugin, and third plugin may be the same plugin. Alternatively, each of the first plugin, second plugin, and third plugin may be different plugins.

[0120] While the method is discussed in reference to a first plugin, a second plugin, and in some embodiments, a third plugin, various embodiments of the present disclosure may include any number of plugins included in a flow with each plugin included in the flow portion of the flow designer interface.

[0121] In various embodiments, the flow portion of the flow designer interface may connections between the renderings of the various plugins, such that the flow of data is shown between plugins. As such, the representation of the third plugin is connected to at least one of the representation of the first plugin or the representation of the second plugin based on the third plugin input.

[0122] Referring now to Block **318** of FIG. **3B**, the method includes generating a flow operation based on the flow portion of the flow designer interface. In various embodiments, a flow operation may also be called a flow. The flow may include each of the plugins represented in the flow portion of the flow designer interface. As such, generated flow operation comprises one or more operations to execute the flow designed in the flow portion (e.g., the data may be processed by one plugin in the flow portion and then transmitted to another plugin in the flow portion, and so on).

[0123] As each plugin is individually pre-coded, the flow generated consists of operations in which the plugins interact with one another. For example, data sets may be received by a first plugin, processed and output to the second plugin to be processed. In various embodiments, the flow operation generated includes each of the plugins represented in the flow portion of flow designer interface. For example, in an instance in which the first plugin and second plugin are represented in the flow portion of flow designer interface, the flow generated includes each of the first plugin and second plugin.

[0124] Referring now to optional Block **320** of FIG. **3B**, the method includes causing an execution of the flow operation generated based on the flow portion of the flow designer interface. In various embodiments, a flow may be executed using the various examples discussed herein, such as via the scheduler. As the generated flow operation discussed in reference to Block **318**, the flow operations include the code necessary to carry out each of the plugin operations. As such, the system is configured to execute the flow as designed.

[0125] Various operations of the system are detailed in reference to FIGS. **4-18**. Each of the operations may be included in the method discussed above in reference to FIGS. **3A** and **3B**. Various embodiments of the present disclosure provide a web-based no-code visual development platform via a flow designer interface that allows complex data compilation, manipulating, and/or the like without requiring complex coding skills. Various embodiments of the platform include data ingestion, data quality checks, data rules, flow creation and modification, flow execution scheduling, calculation integration, and/or various other features into an interactive environment (e.g., a flow designer interface) that provides an end-to-end solution from determining data sources to reporting and analytics. The system also supports data enrichment, data transformation, process automation, and/or the like.

[0126] In various embodiments, the system may include a data ingestion layer of the platform. The data ingestion layer allows configuration of a driven ingestion layer in which the connection parameters to connect with data sources of various data store types may be configured. As such, the system may be capable of loading a large number of rows of data.

[0127] In various embodiments, the system may generate a graphical user interface (e.g., part of a flow designer interface discussed herein) with a data feeds interface may be useful to define metadata about different types of input data which the system can accept. In various embodiments, the system may receive an input from the end-point device that defines the file/table structure, details about actual file and location where files are available to pull. Data can be read into the system through a file feed (FIG. **4**), SQL loader (FIG. **5**), and/or through other workflows.

[0128] As shown in FIG. **4**, the feed builder **400** may include information relating to the data feed (e.g., the source of the data). The data feed may be selected by a user (e.g., from one or more potential data feeds) or entered by the user (e.g., the user may input information relating to the data feed). The metadata page **405** of the feed builder **400** may include metadata relating to the data feed, such as feed name **410**, source name **415**, file pattern **420**, file extension **425**, trigger pattern **430**, trigger extension **435**, field separator **440**, and/or previous inputs **445** received from the feed and/or source. In various embodiments, the selection may be changed by a user or fixed by the system. FIG. **5** illustrates another example metadata page **510** of feed builder **500** in which the feed is using an SQL loader. As such, the metadata page may include feed name **410** and source name **415**, just as the metadata page **410** shown in FIG. **4**. However, instead of file pattern, the feed details may include a selection for the query to pull data **510**. As such, the query used for the SQL loader is shown.

[0129] Referring now to FIG. **6**, the system may also include data dictionary functionality, which is capable of knowing the state of data within the flows discussed herein. The system is capable of determining the exact state of any data at any point in a flow process. In order to determine the data definition of any data, the system may determine two attributes for the data. Namely, the system determines the data type and value of the data. Any changes to the attributes are monitored at each step along the flow. To do this, the initial definition of data type attribute is assigned automatically by the system based on the data, with additional ability for the user to modify the assigned type in the feed configuration module. The data type is shown in Column **610** of the feed builder **600** shown in FIG. **6**. Example data types may include string, integer, decimal, Boolean, date, timestamp, string array, number array, and/or the like. Additional information included in the data dictionary section **605** of the feed builder may include the field name, indexable (e.g., true/false) columns, required (e.g., true/false) columns, and/or the like.

[0130] As data is changed and/or updated by different plugins during a flow, the system may evaluate the data dictionary before and/or after each plugin is executed on the flow based on the user configurations to update the data dictionary. The data dictionary may be continuously evaluated. At each plugin of the flow, the data may have new fields added and/or existing fields modified or deleted, making the traceability of the fields transparent in the system.

[0131] In various embodiments, the system may source data from different upstream sources for transformations to derive new data. The data loading in the feed module is configured with an initial static data dictionary and sent to the flows for use. As such, the data dictionary is defined before the flow is executed and then updated during execution of the flow. Additionally, in an instance in which data is adjusted and/or enriched, as discussed herein, and new data is added to the data sets, the data dictionary may be updated to include the new data added. For example, an enricher plugin may add a new column to a data set and the data dictionary may be updated to include the data definition for one or more values added to the data.

[0132] In various embodiments, the system may include a flow designer that is configured to organize the operations of the system herein. For example, the flow designer is configured to perform operations relating to workflows, rules, calculations, reports, analytics and/or the like. In various embodiments, the system may include a flow designer interface that includes individual plugins that each perform a transformation on the data in the flow. The flow designer interface may allow multiple flows, where each flow is an individual transformation with a flow input and a flow output. Each flow is independent from the other flow, though in some instances, the output of a flow may be used in other flows. An example flow may consist of determining data sources, data transformation, data writing, data reporting steps, and/or the like.

[0133] In various embodiments, the flow designer interface represents each plugin as a block and allows for the blocks to be visually connected to one another to represent the transformations being completed. An example flow designer interface is shown in FIG. 7. The flow designer interface **700** may include a plugin portion **705** that includes available plugins that can be used in the flow portion **725**. As shown, the plugins may be represented in the flow as blocks (e.g., blocks **701**, **702**, **703**, **704**, and **710**). As such, each flow may be connected to one another (e.g., as shown Block **702** leads to Block **703**, which leads to both Blocks **704** and **710**). Within each plugin users can define specific rules, logic, filters, provide data enrichments, data transformation, complex calculations, user defined fields, aggregate data, and/or the like.

[0134] In various embodiments, the system may include an end-to-end modularized data platform that enables users to digitize and optimize processes and workflows. To do this, the system combines data ingestion, data quality rules, business rules, workflows, calculation integration, testing, and/or the like, into maneuverable design environments. Additionally, the system provides an end-to-end solution from determining data sources to reporting & analytics.

[0135] In various embodiments, the system may include a flow designer assistance via AI functionality. As such, the system monitors users during operation of the flow designer platform and may provide recommendations and/or corrections to the actions by the user. For example, the system may automatically correct workflow steps in an instance in which a change is made in any plugin and/or provide suggestions for improving plugin configuration when any config changes are made. As such, the system allows for a user to make required changes at all appropriate places without having to manually scan through the configuration of every single plugin.

[0136] In various embodiments, the flow designer allows for plugins to be easily moved, added, removed, and/or changed in the system. However, the AI functionality allows for safeguards to protect the underlying data from being damaged and/or manipulated incorrectly. As such, the system is configured to monitor changes to the flow designer for functionality. For example, a user who has developed a process through 40 individual steps using 8 different data inputs on the workflow designer may want to make changes in steps 12 and 18 along with changes to the file structure for 2 input files. The system scans each workflow component (e.g., plugins) to determine

which workflow components are affected, thereby reducing the time required to make adjustments (e.g., typically, the user would be required to manually scan all 40 individual workflow steps to identify which steps would be impacted by these changes). The system may provide prompts to the user via the platform interface (e.g., via a graphical user interface) that assist the user in updating any workflow components that are affected by the change in the workflow.

[0137] Referring now to FIG. 7, the system may include one or more flow output writers that are used to break down processes into smaller, more understandable processes. The flow output writer (e.g., Block **710** of the flow designer interface **700**) allows for the output of a flow available to be an input/feed to be used in other flows. To do this, the system generates the output of a plugin at a run time and creates an automatic data feed which can be used as an input into the next flow. As such, the output of flows may be used in other flows, as a single block of the flow, allowing multiple flows to be used together without requiring each individual step of the flow to be manually entered.

[0138] In various embodiments, the system may also include an embedded flow plugin that allows for the flow to behave like a custom function where arguments can be changed and executed for different use cases. As such, the system may allow for the input of the flow to be changed, while maintaining the operations of the flow to remain the same, allowing the flow to be executed on different data inputs. The embedded flow is unique as the system allows for a flow to be reused and/or repurposed using different inputs with the same flow operations. As such, a flow may be represented as a single plugin to be used as an embedded flow within another flow.

[0139] Referring now to FIG. 8, an example of flow designer interface **800** is shown using a multi-level grouping functionality of various embodiments. Since flows may include large number of individual operations (e.g., plugins that make up the flow), the system may use grouping functionality in order to allow for the entire flow to be represented on a single flow designer interface. As such, each group may include one or more operations in which the data is being transformed. As such, the flow may include a first group (e.g., Group 1 **850**), a second group (e.g., Group 2 **855**), and a third group (e.g., Group 3 **860**). The individual groups may be connected to one another. For example, as shown in FIG. 8, Group 1 **850** is part of Group 2 **855**, and Group 2 receives an input from Group 3 **860**. The flow designer interface **800** allows for the groups to be minimized or expanded. For example, Group 1 **850** is minimized and does not show the operations therein, while Groups 2 **855** and Group 3 **860** are expanded and show the individual plugins. As such, the groups act as a plugin in the flow designer interface. In various embodiments, the groups may also be connected to other plugins. As shown, Group 2 is an input for Block **865**. In various embodiments, the groups may be customized, such as a customizing the name of the group, to provide additional information relating to the group.

[0140] In various embodiments, multi-level grouping allows for the flow designer interface to have a simplified visual display that allows for users to view large numbers of operations within a single interface.

[0141] Referring now to FIGS. 9A-9B, an adjustment plugin is shown, including how the adjustment plugin may be placed into a flow designer interface and the adjustment plugin may be adjusted by the user. FIG. 9A shows the adjustment plugin **910** placed on the flow designer interface **900**. As shown, input from a repository **905** may be adjusted via the adjustment plugin **910**. The data may then be enriched via the enricher plugin **915** after the data adjustment. The adjustment **910** may be modified via the adjustment portion **920** of the flow designer interface **900**. The adjustment portion **920** may include the type of adjustments made to the data during the adjustment. For example, the input from a repository **905** may be adjusted by adding new rows (e.g., Add_Rows **925**), filtering out some rows (e.g., Filter_Rows **930**), and Updating some columns (e.g., Update_column **935**). In various embodiments, the adjustment plugin allows for data to be adjusted after data has been manipulated and/or compiled. As shown in FIG. 9B, the adjustments made in the adjustment plugin may be changed via the flow designer interface **900**. As

shown, the adjustment type **950** may be changes, as well as the adjustment name **955**, and the portion of the data being adjusted (e.g., HLD_PRD_83 **960**).

[0142] In various embodiments, different types of adjustments may be made to data in the flow. Example adjustment types include create (e.g., add rows and/or columns), filter (e.g., remove columns and/or rows), and/or update (e.g., change values in data). The adjustment plugin allows for adjustments to be applied at any step within the executed workflows in order to modify the calculated output dataset. The adjustment plugin may require multiple approvals before being implemented into a flow. For example, the system may receive an adjustment request (e.g., via a first user, such as an analyst), and an adjustment approval from an authorized approver (e.g., a second user, such as a manager) before the adjustment plugin is used in the flow. In various embodiments, the system may allow the adjustment to be toggled between applied and not applied. As such, the data may be viewable in the flow designer platform as both adjusted and not adjusted by a user.

[0143] In various embodiments, the system allows users to adjust existing flow components (e.g., plugins) on an executed workflow leveraging pre-defined adjustment types. As such, the adjustment plugin allows for the flexibility to introduce changes to the data flows, while preserving the integrity of the data. Example of adjustments include inserting new rows of data, updating sets of data by defining conditions for one or multiple columns meeting specific analytics requirements, filtering or removing unwanted rows or columns of data, and/or the like. In various embodiments, one or more adjustments may be deactivated and activated within the flow designer interface, allowing adjusted data to be reverted to an unadjusted state.

[0144] In various embodiments, the adjustment plugin supports high volumes of data, complex data transformation, and real-time analytics changes using one or multiple plugins.

[0145] Referring now to FIGS. **10** and **11**, the system may also provide an ability for users to input custom code for one or more plugins in the flow. While the plugins are each able to be used without needing any code, some users may also want to add new or otherwise non-supported functionality that are not covered by the plugins offered by the system. As such, the system may provide a code entry system and the system may also include code verification in order to assure code operability. In various embodiments, the system may include an advanced, snippet function in which custom code may be used in a flow.

[0146] Snippet function allows for users to create custom snippet plugins via custom code. The custom plugins may then be available across the flow network (e.g., other network users may be able to use the custom plugin in the given user's own flows). In various embodiments, the code created using the Snippet function can be made viewable to all network users or viewable to select network users. For example, a network user may be able to use a custom snippet plugin without having access to the code of the custom snippet plugin. As such, the custom snippet plugin may operate just as other plugins for a network user (e.g., the custom snippet may be a block in the plugin portion of the flow designer that can be added to a flow). Additionally, the custom snippet plugins may be used in multiple different plugins, allowing a network to customize the flow designer interface based on the specific use of the network (e.g., an entity may use the flow designer for a specific use case).

[0147] As shown in FIG. **10**, the snippet function may have a specific page **1000** of the flow designer interface that allows for the snippet code to be added via the snippet code portion **1005** of the interface and assigned as a custom snippet plugin. The system may support various different computing languages, such as Java, Python, R, SQL, and/or the like. In various embodiments, the language may be autodetected or selectable by the inputting user (e.g., the Select Language **1010** is JAVA in FIG. **10**). The system may include coding verification and/or review that allows the code to be approved before being used as a plugin.

[0148] FIG. **11** illustrates a custom snippet plugin being used in a flow. As shown, the custom snippet plugin **1105** may be used as a regular plugin in a flow designer interface (e.g., flow

designer interface **1100**). As shown in the JAVA snippet portion **1110** of the flow designer interface, information relating to the custom snippet plugin may be displayed, such as selecting which snippet to use (e.g., the same snippet is “NettingSet id 1” is selected as the custom snippet plugin is named in FIG. **10**). As such, the custom snippet plugin may be used with no requirements for programming languages of other plugins in the flow.

[0149] In various embodiments, the system includes an enricher plugin tool. The enricher allows a user to carry out data manipulation tasks across one or more records. The data manipulations may include any function used to modify data. For example, the data manipulation may range from simple mathematical functions, string-based manipulation, if/else style conditional checks, and/or the like. Various different functions of the enricher may be provided via a graphical user interface. In various embodiments, the enricher may also include the functionality of the adjustment plugin discussed herein.

[0150] In various embodiments, the enricher allows for a user to take a data set that has been created (e.g., via the data flow designer discussed herein) to be adjusted without requiring the data to be re-processed. For example, a user may want to add new columns or other functions to a data set. The enricher provides this functionality with minimal to no coding requirements.

[0151] In various embodiments, the system may include one or more rule engines that may be designated via the flow designer interface. Example rule engines include data quality rules, transformation rules, statements rules, truth tables, matrix rules, sequential rules, and/or various other rule engines. As such, the rule engines may be used as individual plugins in the flow designer interface.

[0152] Referring now to FIGS. **12A-12C**, various user interfaces are shown to generate a data quality rule. In various embodiments, the system may include data quality rules. The data quality rules may be used to identify data issues before and/or during the flow execution. The data quality rules may be automated. As such, data quality rules may be defined for a particular field in a dataset or aggregated for the entire feed. The system, via the data quality rules, may test whether the data set meets the defined criteria, whether a data set contains data in required fields, and/or the like. As such, the system may determine one or more issues relating to the data set.

[0153] In various embodiments, the system may use the data quality rules to monitor data accuracy (e.g., the accuracy of data in fields), data validity (e.g., checking whether the values are in specific format or not), data completeness (e.g., completeness of values and checking if all data is present), data transmission timeliness (e.g., timeliness of files arrival), consistency of data (e.g., checking the duplicates and consistency), data integrity (e.g., checking whether reference values are populated as expected or not), and/or the like. As shown in FIG. **12A**, a data quality rule generator page **1200** may be included in the flow designer interface to allow for data quality rules to be defined. The rule category may be entered by the user (e.g., entered manually or selected from a drop-down menu, such as a selection menu **1205**).

[0154] In various embodiments, data quality rules may be created at the feed level and/or the flow level discussed herein. In various embodiments, data quality rules may be created at feed level before file execution. As such, the system can run the data quality rules once the feed is received and identify data anomalies. In various embodiments, the data quality rules may be created, modified, and/or the like via an end-point device. Example types of data quality rules that may be created include simple column validation, cross column validation, feed level validation, and/or the like. As shown in FIG. **12B**, the rule type may also be input by the user (e.g., entered manually or selected from a drop-down menu, such as selection menu **1210**). The rule type selection may be included in the data quality rule generator page **1200**, as shown in FIG. **12C**. Additionally, as shown in FIG. **12C**, the rule specification may be input by the user (e.g., entered manually or selected from a drop-down menu, such as selection menu **1215**). In various embodiments, the system may generate one or more of the rule category, rule type, and/or rule specification, such that a user does not have to input the information (e.g., the system may have a predefined selection that

may be changed by the user or a predefined selection that is not changeable by the user).

[0155] In various embodiments, a simple column validation may be a data quality rule used to validate the data for one field in a feed. Simple column validation may check for string pattern, date pattern, custom SQL, required field values, minimum/maximum value, unique values, and/or the like.

[0156] In various embodiments, a cross column validation may be a data quality rule used to validate the data for a combination of fields in a feed. Each row of a data set may be validated based on the data quality rules for combination of fields. For example, one can check if there are duplicates based on combination of col1, col2 and col3. Cross column validation may check for unique values, custom SQL, and/or the like.

[0157] In various embodiments, a feed level validation may be a data quality rule used to validate data at feed level. For feed level validation, a single result output may be generated for the entire feed. For example, the system may check if the total number of records in a feed is within 5 percent of 1 million. In various embodiments, the system may be able to filter the data set to perform feed level validation. For example, the system may check the threshold limits for absolute value or based on last output. Feed level validation may check for total number of records, sum of data sets, average of data sets, custom SQL, and/or the like.

[0158] Referring now to FIG. 13, the system may cause a rendering of a user interface with results of the data quality rule monitoring. For example, the flow designer interface may include a data quality dashboard **1300** that includes the results of one or more data quality rule results. For example, the data quality dashboard may include which data quality rules passed on a data set and information relating to the data set, such as the percentage of the data set that passed the data quality rules.

[0159] The data quality dashboard, such as the data quality dashboard **1300** shown in FIG. 13, provides an overall picture about the quality of data at the feed level for each feed. In the dashboard, a user associated with an end-point device can see various measures, such as type of data quality rule (e.g., Validity, Consistency, Accuracy, Integrity, Completeness, Timeliness), percentage of good records for each field, and/or the like. Railed records may also be downloadable in order to do further analysis. For example, the data quality dashboard **1300** may include the overall validity of the data **1305**, the number of records scanned **1310**, the number of data quality rules checked **1315**, the individual data quality for each data set (e.g., via the attribute summary section **1320**). For example, for the data file named “Regime”, the data quality is mostly good quality (e.g., represented by bar **1325**) with only a small subset that is considered bad (e.g., represented by bar **1330**). In various embodiments, the data quality dashboard **1300** may also include an insights section **1335** that includes anomalies detected in the data sets.

[0160] Once the feed level data quality rules are executed, the data quality dashboard may provide the overall quality of data. The system may make decisions before the data is consumed.

Additionally or alternatively, a user may make decisions relating to the data. In an instance in which the data is determined to not be good enough, then actions may be executed, either automatically or manually, such as reaching out to the corresponding source team. As such, delays in file delivery of a feed may be identified. In an instance in which the data quality rule output executed at flow level data has predetermined number of bad records, the source data may be reviewed, and appropriate actions may be taken (e.g., updating the source data). As such, the data quality dashboard provides a powerful adjustment management feature that is easy to visualize, understand, design, and debug.

[0161] Data quality rules may be generated for each plugin within a flow. As such, the data quality rules may be created for the feed level and/or the flow level. As such, the data quality rules may be executed on individual feeds or across the entire flow.

[0162] In various embodiments, data quality rules allow the system to identify potential data issues by various pre-defined rules. As such, the system may detect potential weak points in processes and

generate recommendations for action. Based on the DQ rule output, the system may determine that the data needs undergo cleansing and enrichment. For example, duplicates in the data may be removed, data for missing elements may be added, and/or the like.

[0163] In various embodiments, the system may include a scheduler that allows for workflow(s) to be scheduled for execution. The scheduler allows for execution to be scheduled at a pre-determined time or when a data feed/file is ready to be consumed or triggered from an upstream system. As such, the scheduler may have triggers to execute different flows.

[0164] In various embodiments, triggers for flow execution may include time-based triggers, file-based triggers, and/or manual-based triggers. In time-based triggers, the same business workflows can be made to execute under different times with different input parameters that are required for the workflows using the over-ride functionality in the scheduler. In file-based triggers, flows are executed upon arrival of at least one data file from upstream and can also be set on multiple files. In manual-based triggers, the flows are executed based on the pre-set configurations and are triggered by the user. The user can modify the input parameters before execution. This trigger also allows the user to execute a flow for days in the past.

[0165] Referring now to FIGS. **14A-14D**, creation of a scheduled execution in the scheduler via a graphical user interface. As shown, in FIG. **14A**, a flow execution scheduler interface **1400** may include an execution name **1405** and a trigger type **1410** (e.g., time-based, file trigger, manual trigger, etc.). The flow execution scheduler interface **1400** may also include a variable override portion **1415** in which one or more variables can be overridden (e.g., deactivated in an instance the variable being created is used). The graphical user interface may also request additional information relating to the trigger (e.g., FIG. **14B**). For example, a time-based trigger may include the time of trigger and information relating to recurring execution as shown by selection **1420** (e.g., the execution may be repeated daily, weekly, monthly, etc.). FIGS. **14C** and **14D** illustrate different trigger types (e.g., file trigger in FIG. **14C** and manual trigger in FIG. **14D**). For the file trigger type shown in FIG. **14C**, the flow execution scheduler interface **1400** may include a trigger selection portion **1425**, in which the different triggers may be defined and/or selected. As shown, some triggers may be selected and/or deselected for a given variable. For a manual trigger, such as the one shown in FIG. **14D**, the flow execution scheduler interface **1400** may include the process time **1430**, which indicates the time at which the flow is to be triggered. The manual trigger and file triggers may also include the ability to override variables.

[0166] After initial configuration of a flow execution is provided, the flow execution may be automated without any additional information. For example, the time-based and file-based schedulers may be automatically executed upon being triggered. The system may be capable of running the same flow executions on multiple schedules with different input parameters through the over-ride functionality within the system.

[0167] In various embodiments, a flow is a collection of steps and rules used to automate a data process. In various embodiments, the data flow may be generated by a user via the flow designer discussed herein. For example, a typical flow may consist of determining data sources, data transformation, data writing and data reporting steps. The scheduler discussed herein allows workflows to be automated without the need of manual intervention after the initial configurations are completed. The workflows can be executed using the scheduler at a predetermined time (e.g., daily at 10:00 AM), based on the arrival of any upstream data files or from a database call, and/or manually by a user.

[0168] In various embodiments, the system may cause rendering of an execution dashboard that provides transparency of end-to-end flow execution along with rules, calculations, custom plug-ins, the ability to draw lineage for each data operation, and/or the like. FIG. **15** shows an example execution dashboard **1500**. The execution dashboard allows a user to quickly view the status of workflow execution and start utilizing processed data for downstream reporting. The execution dashboard may include a variety of statistics such as time taken, data quality issues, data lineage,

status of overall process, and/or the like. Details of each execution also gives lineage details for each data point, row or column of data.

[0169] As shown in FIG. 15, the execution dashboard **1500** may include a list of one or more flows (e.g., each row of the chart shown in FIG. 15 is a flow) with each flow including information relating to scheduled execution, previous executions, last execution status message, progress of last execution, last execution start time, last execution end time, cause of last execution, and/or the like. For example, for the flow named “Counterparty flow_RM_4” **1505**, execution is scheduled daily at 5 AM. The last four executions have included three successful executions (indicated by the double check mark) and one unsuccessful executions (indicated by the Exclamation point). The last execution was completed in 2 second and was 100% completed (as indicated by the progress bar **1510**). Meanwhile, the last execution of Flow 13 **1515** failed at 44% completion.

[0170] Referring now to FIGS. 16A-16D, the system may include data tracing that can be used within the flow designer interface. The data tracing feature allows users to trace any individual data row or individual data column in both directions through a flow (e.g., from input to output tracing and/or from output to input tracing). As such, the system allows network users to easily understand how each individual data point is sourced, how the data point has been derived and transformed at each step of the flow through a visual representation. Data tracing allows for the individual rows, columns, and/or cells to be traced for each plugin. As such, the system provides a visual representation of the input and output data sets for each plugin.

[0171] Data tracing allows users to review data at each step of the process in both directions. Forward tracing from input to output and backward tracing from output to input. The system may include data tracing performed at the column, row, and/or cell level.

[0172] Column tracing gives the users the ability to see where in the flow a column is used and allow for the any transformations on the column to be located. For example, in an instance in which a column is traced, the system may highlight all the plugins that use that column on the flow designer interface. In an instance in which a pivot column is traced, the system may highlight the plugins where the original column occurs.

[0173] Row tracing allows network users to follow the directions of the data. The row can be traced in the same or multiple workstreams. The system may display various data transformations and enhancement, aggregated fields, inspect filters, joins and various data enrichments, and/or the like. In various embodiments, the system may allow data tracing a row from an aggregation or a pivot plugin. In such an example, the system may highlight all the rows that compile that aggregated row or pivoted row.

[0174] Data cell tracing is a combination of row and column tracing. As such, the system traces a data cell, by highlighting all the plugins in the workflow where that given row and column of the data cell appears. Users may then be able to click on each highlighted plugin and view the data.

[0175] During data tracing, the system may enter inspection mode in which data manipulation is deactivated, such that the data may be traced without any unwanted changes to the data sets. At each step of data tracing, the system may allow the data to be exported (e.g., to local folders in a .csv file format).

[0176] In various embodiments, data tracing may be carried out on a flow that has previously been executed successfully. Execution dashboard (e.g., execution dashboard **1500** shown in FIG. 15) allows a user to quickly view the status of workflow execution and start utilizing processed data for downstream reporting.

[0177] Referring now to FIG. 16A, data from any plugins in a flow may be data traced. As shown, plugin **1605** is selected and the data set is shown in the data portion **1610** of the flow designer interface **1600** (via selecting icon **1615**). As shown, a network user may select the type of data tracing (e.g., here, the user selects Trace Row **1620**). FIG. 16B illustrates an example flow designer interface **1600** in which inspection mode has been activated. As discussed above, inspection mode prevents the data from being manipulated, but allows user to use the data tracing feature. As shown,

the row **1650** is being data traced. Each of the plugins that are part of transforming the data in the row are shown as highlighted blocks, while the other plugins are greyed out. As shown in FIG. **16C**, the row originated from an upstream flow (e.g., another flow in which the plugin **1675** is represented. As such, the system may allow the data tracing to continue to the previous flow (e.g., FIG. **16D** illustrates the flow represented by plugin **1675** in FIG. **16C**).

[0178] In various embodiments, the system may define variables to be used during flow execution. Examples of variables defined on the system include global variables, flow variables, and system variables. Global and flow variables may be defined by system users. System variables may be a set of predefined read-only dynamic variables (e.g., system date, user id, etc.) available to all the applications onboarded on the network. Global variables are available to all the flows, while the flow variables are available only in the flow it is defined and any child flows.

[0179] Global and/or flow variables allow for flows to be re-purposed and/or re-used using different variable. For example, the same flow can be used with different global variables in order to produce two different, but desired data outputs. For example, some flows may use the same data feeds but operate on only a subset of the data for each flow and variables allow a single flow to be used instead of multiple flows. Global and/or flow variables may be defined as static or dynamic. In an instance in which a variable is static, a user may provide the values of the variable that remain constant during execution of the flows. In an instance in which a variable is dynamic, a system may receive a configuration from a user for the variable, and the system may then determine the values to be used as the variable during run time on the day of execution. Dynamic type variables may be generated using various generators such as a date generator, a sequence generator, a SQL generator, and/or the like.

[0180] Referring now to FIG. **17**, a global variable generation interface **1700** for defining a global variable in the system is shown. The global variable generation interface **1700** may be part of the flow designer interface discussed herein. As shown, the interface may allow various different components of the global variable to be defined. While FIG. **17** illustrates a global variation generation, the same operations may be used for flow variables with the flow variables only applying to a specific flow.

[0181] In various embodiments, the global variable generation interface **1700** may input selection for the global variable name **1705**, global variable description **1710**, select generator **1715**, select data type **1720**, global variable value **1725**, select application **1730**, and/or the like. Example select generator **1715** options include date generator, sequence generator, SQL generator, and/or the like. The select generator may be indicated as none in an instance in which the variable is static. Example select data type include string, integer, decimal, Boolean, date, timestamp, string array, number array, and/or the like. In various embodiments, the variables allow for the flows to be adapted and/or changed for different conditions without changing any of the actual operations of the flow.

[0182] In various embodiments, the system may generate analytics and reports based on the flow executions discussed herein. An example data quality dashboard is shown in FIG. **13** and discussed above. An example analytics dashboard is shown in FIG. **18**. The analytics dashboard **1800** may include analytics (e.g., statistics relating to the flow executions), reports generated relating to the flow executions, and/or what-if scenarios (e.g., simulated flow executions). As such, the system may provide access to real-time or near real-time information via the analytics dashboard **1800**. In various embodiments, the system may create end user reports and analytics to be displayed on the analytics dashboard **1800**.

[0183] As shown in FIG. **18**, the analytics dashboard **1800** may include various different dashboard components, such as data quality component **1805**, an input data summary **1840**, a feed load status summary **1845**, a flow builder summary **1850**, a flow execution summary **1855**, and summary of analytics for time periods (e.g., today's activity **1860** and/or month's activity **1865**).

[0184] The data quality component **1805** may include an accuracy indicator **1810**, a completeness

indicator **1815**, a consistency indicator **1820**, a timeliness indicator **1825**, a validity indicator **1830**, a uniqueness indicator **1835**, and/or the like. Each of the indicators on the data quality component **1805** may relate to one or more data sets in one or more flows. As such, the percentage shown in each indicator may correspond to all of the data sets in the one or more flows.

[0185] The input data summary **1840** may include a chart of the percentages of different input data sets. As shown, the input data in the example has four different input data (e.g., input data 1, input data 2, input data 3, and input data 4). The feed load status summary **1845** may include information relating to the amount of the data received from the input sources. For example, as shown, 58% of the total input data may be received by the system.

[0186] The flow builder summary **1850** may include the total flows built, the flows that were executed, and the flows that are created, but not yet executed (e.g., draft flows). The flow builder summary **1850** may include a chart relating to the data. The flow execution summary **1855** may also include visual representations of the flow executions.

[0187] In various embodiments, analytics based on time periods may be displayed on the execution dashboard **1800**. For example, today's activity **1860** may include various different statistics relating to the flow executions for a given day. Additional time periods may also have a dashboard component. For example, the month's activity may also include information relating to the flow executions for the given month.

[0188] FIG. **19** includes a flow chart **1900** that illustrates an example method of determining and maintaining data quality in data processing. The method may be carried out by various components of the distributed computing environment **100** discussed herein (e.g., the system **130**, one or more end-point devices **140**, etc.). An example system may include at least one non-transitory storage device and at least one processing device coupled to the at least one non-transitory storage device. In such an embodiment, the at least one processing device is configured to carry out the method discussed herein. A method of various embodiments may include any combination or subset of the features discussed herein.

[0189] Various operations of the system are detailed in reference to FIGS. **3-18**. Each of the operations may be included in the method discussed in reference to FIG. **19**. Various embodiments of the present disclosure provide a web-based no-code visual development platform via a flow designer interface that allows complex data compilation, manipulating, and/or the like without requiring complex coding skills. Various embodiments of the platform include data ingestion, data quality checks, data rules, flow creation and modification, flow execution scheduling, calculation integration, and/or various other features into an interactive environment (e.g., a flow designer interface) that provides an end-to-end solution from determining data sources to reporting and analytics. The system also supports data enrichment, data transformation, process automation, and/or the like. Specifically, the data quality rules discussed in reference to FIGS. **12A-12C** may be used in the various operations discussed in reference to FIG. **19**.

[0190] Referring now to Block **1902** of FIG. **19**, the method includes determining one or more plugins used in a flow execution. As discussed herein, a flow execution may be a flow as discussed herein that is executed. In various embodiments, a flow execution may include performs at least one transformation on one or more data sets. The various plugins may be the same plugins that are shown in the flow portion of the flow designer interface. In various embodiments, the data quality rules discussed herein may be used on flows that are generated outside of the flow designer interface as well.

[0191] In various embodiments, in an instance in which the flow was generated using the flow designer interface, the plugins may be determined based on the flow portion of the flow designer interface (e.g., the representations on the flow portion indicate the plugins in the flow). The system may determine and/or receive various other information relating to the plugins in the flow, such as plugin type (e.g., transformation, data source, etc.), order of data flow relating to the given plugin (e.g., the input received by a plugin and other plugins that receive the output of the given plugin),

typical plugin operations (e.g., example output of the plugin, average execution time, etc.) and/or the like.

[0192] Referring now to Block **1904** of FIG. **19**, the method includes determining one or more automatic data quality rules based on at least one of the one or more plugins using in the flow execution. An automatic data quality rule is a data quality rule that is predetermined to be used for a flow and/or feed. The data quality rule may any of the data quality rules discussed above in reference to FIGS. **12A-12C**.

[0193] Each automatic data quality rule may be assigned data to monitor, such as the input (e.g., from a data source), an intermediate input and/or output (e.g., the input and/or output for a given plugin in the flow), the output of the flow, and/or the like. As such, the same data quality rule may be used for different data associated with the flow (e.g., the uniqueness of the data may be monitored for each data set output by a plugin in the flow).

[0194] In various embodiments, the system may determine the automatic data quality rules based on the plugins used in the flow. For example, certain plugins and/or plugin types may be assigned certain data quality rules frequently and the system may then automatically use said data quality rule in future flows with the same plugin. As such, the automatic data quality rules may be selected based on historical data quality rules across previously generated and/or executed flows. In various embodiments, the AI/ML model(s) discussed herein may be trained to selected automatic data quality rules. For example, the AI/ML models may be trained using one or more flows that have data quality rules assigned to the given flow. As such, the automatic data quality rules may be assigned based on similarities between the flow and other flows (e.g., a flow with a combination of certain plugins may have the same data quality rules automatically assigned).

[0195] In various embodiments, one or more automatic data quality rules may be assigned to each flow. For example, a network administrator or the like may decide that each flow should monitor the process speed of the flow and the process speed of the flow is then selected as an automatic data quality rule for each flow.

[0196] In various embodiments, one or more of the automatic data quality may be selected based on a selection by a network user relating to the automatic data quality. For example, the system may provide a rendering of an interface in which a network user may select one or more of the automatic data quality rules. In such an embodiments, the flow and/or feeds in which the automatic data quality rules are to be applied may also be selected (e.g., a network administrator may indicate that a given data quality rule should be used any time a specific plugin is used in a flow).

[0197] In various embodiments, one or more of the automatic data quality may be selected based on a custom data quality rule for a previous flow. For example, during the creation of the custom data quality rule(s) discussed herein, a network user may be able to indicate whether the custom data quality rule should be used with future flows and/or feeds. As such, in an instance in which the network user indicates that the custom data quality rule should be used with future flows and/or feeds, the given data quality rule may be selected as an automatic data quality rule for a future flow. In various embodiments, custom data quality rule may automatically be considered to be selected as an automatic data quality rule in future flows and/or feeds. For example, the AI/ML models may be trained using the flow execution, such that the given custom data quality rule may be selected as an automatic data quality rule for similar flows in the future (e.g., flows with one or more of the same plugins used).

[0198] Referring now to optional Block **1906** of FIG. **19**, the method includes causing a rendering of a data quality rule interface configured to receive the custom data quality rule input. In various embodiments, the data quality rule interface may be part of the flow designer interface discussed herein. As such, the data quality rule interface may be part of the flow creation and/or modification (e.g., a data quality rule can be added for a given flow).

[0199] In various embodiments, the platform discussed herein may include the data quality rule interface as a page caused to be rendered on an end-point device. In various embodiments, the data

quality rule interface may include information relating to the one or more automatic data quality rules, previously created custom data quality rule(s), predefined data quality rules that may be used as custom data quality rules and/or the like.

[0200] The data quality rule generator page **1200** shown in FIGS. **12A-12C** may be included in the data quality rule interface, such that the entering of information into the data quality rule generator page is a custom data quality rule input. The system may determine the parameters of the custom data quality rule based on the custom data quality input. In various embodiments, the system may have one or more predefined parameters, such that the network user does not have to provide all of the information relating to the custom data quality rule. For example, the rule category may be predefined (or at least suggested) based upon other entries by the network user. In various embodiments, the predefined parameters may be changeable by the network user.

[0201] Referring now to optional Block **1908** of FIG. **19**, the method includes removing at least one of the one or more automatic data quality rules based on an input from an end-point device. In various embodiments, the one or more automatic data quality rules may be edited by a user. The editing of the automatic data quality rule(s) may be done at the same or similar time in which the custom data quality rule(s) are created (e.g., a network user may be provided a data quality rules interface in which the network user can edit the automatic data quality rule(s) and/or create custom data quality rule(s)). As such, the removal of at least one of the one or more automatic data quality rules may be based on a user input received from an end-point device associated with a network user.

[0202] In various embodiments, the automatic data quality rule(s) may be provided to the user on the end-point device (e.g., via the data quality rule interface) and the network user may select one or more of the automatic data quality rules to deactivate or remove from the automatic data quality rules. In such an instance, the selected data quality rule is no longer considered an automatic data quality rule and is not used for a given flow and/or feed (e.g., the selected data quality rule is not considered an automatic data quality rule for the purposes of the execution of each of the automatic data quality rules and the one or more custom data quality rules on data in the flow execution discussed in reference to Block **1912**). In various embodiments, the selected data quality rule may remain an automatic data quality rule for other flows and/or feeds. Alternatively, the selected data quality rule may be removed from being assigned as an automatic data quality rule for future flows and/or feeds.

[0203] Referring now to Block **1910** of FIG. **19**, the method includes determining one or more custom data quality rules based on a custom data quality rule input received from an end-point device. In various embodiments, the custom data quality rule may include one or more data statistic to be monitored during the flow execution. The custom data quality rule may be any of the data quality rules discussed above in reference to FIGS. **12A-12C**.

[0204] In various embodiments, the custom data quality rule(s) may be generated via a network user inputting information into a custom data quality rule builder (e.g., via the data quality rule interface). As such, the system may prompt the network user for information relating to the data quality rule, such as the data quality rule generator page **1200** shown in FIGS. **12A-12C**. In such an embodiment, the system may provide one or more fields for a network user to input information relating to the custom data quality rule. In some embodiments, one or more of the field(s) may include selections available for the data quality rule (e.g., rule category may include the different rule categories in which the system may be capable of executing).

[0205] In various embodiments, one or more of the custom data quality rules may be selected from one or more predefined data quality rules. For example, some data quality rules may not be selected as automatic data quality rules but are already ready to be implemented in an instance in which the user selects the given data quality rule (e.g., the data quality rule is already defined). As such, the custom data quality rule may be predefined (e.g., by the system, by previous network users, etc.) and selected as a custom data quality rules.

[0206] In various embodiments, the system may recommend one or more predefined data quality rules to be selected as a custom data quality rule. For example, a given data quality rule may be used in similar flows but was not selected as an automatic data quality (e.g., a data quality rule may not be selected as an automatic data quality due to burden of execution, lack of established use (e.g., a data quality rule may be used for only one previous flow that is similar, but was not used in other similar flows), and/or the like. As such, the system may provide one or more predefined data quality rules to the user (e.g., the user may be provided one or more recommended predefined data quality rules via the data quality rule interface.

[0207] As discussed above, a custom data quality rule may be any of the data quality rules discussed above in reference to FIGS. **12A-12C**. For example, each custom data quality rule may include a rule category and the rule category for each of the one or more custom data quality rules may be accuracy based, completeness based, consistency based, timeliness based, or uniqueness based. Each of the data quality rules (automatic and/or custom) may be a column validation, a cross column validation, a feed level validation, and/or the like. For example, a data quality rule may determine whether a data set associated with the flow execution contains a required value.

[0208] Referring now to Block **1912** of FIG. **19**, the method includes causing execution of each of the automatic data quality rules and the one or more custom data quality rules on data in the flow execution. In various embodiments, the execution of the data quality rules may be completed in parallel with the execution of the flow. For example, the data quality rules may be executed during the flow execution as the data on which the data quality rule is being executed becomes available. Additionally or alternatively, the execution of the data quality rules may be after the execution of the flow. As discussed herein, the data at each level of flow execution may be stored, such that the data quality rules may be executed at any time (e.g., the data from the flow including intermediate inputs and outputs are still available in order to execute data quality rules).

[0209] In various embodiments, the various data quality rules may be executed on various data in the flow execution. For example, the data in the flow in which the data quality rules are executed may include the output of the flow execution, an intermediate input and/or output of the flow execution (e.g., an input and/or output of a given plugin in the flow), an input to the flow (e.g., the data received from a data source), and/or the like.

[0210] Referring now to optional Block **1914** of FIG. **19**, the method includes causing a rendering of results for one or more of each of the automatic data quality rules and the one or more custom data quality rules on data in the flow execution to the end-point device. As discussed herein, the system may provide analytics and/or reports for the flows may be generated. The rendering of results may be in the form of a data quality dashboard, such as the data quality dashboard **1300** shown in FIG. **13**. Additionally or alternatively, results of the data quality rules may be shown as a part of additional analytics on the flow, as shown in the analytics dashboard **1800** of FIG. **18**. As such, the rendering of results may be any of the renderings discussed in reference to FIGS. **13** and **18**.

[0211] As will be appreciated by one of ordinary skill in the art, various embodiments of the present disclosure may be embodied as an apparatus (including, for example, a system, a machine, a device, a computer program product, and/or the like), as a method (including, for example, a business process, a computer-implemented process, and/or the like), or as any combination of the foregoing. Accordingly, embodiments of the present disclosure may take the form of an entirely software embodiment (including firmware, resident software, micro-code, and the like), an entirely hardware embodiment, or an embodiment combining software and hardware aspects that may generally be referred to herein as a “system.” Furthermore, embodiments of the present disclosure may take the form of a computer program product that includes a computer-readable storage medium having computer-executable program code portions stored therein. As used herein, a processor may be “configured to” perform a certain function in a variety of ways, including, for example, by having one or more special-purpose circuits perform the functions by executing one or

more computer-executable program code portions embodied in a computer-readable medium, and/or having one or more application-specific circuits perform the function.

[0212] It will be understood that any suitable computer-readable medium may be utilized. The computer-readable medium may include, but is not limited to, a non-transitory computer-readable medium, such as a tangible electronic, magnetic, optical, infrared, electromagnetic, and/or semiconductor system, apparatus, and/or device. For example, in some embodiments, the non-transitory computer-readable medium includes a tangible medium such as a portable computer diskette, a hard disk, a random access memory (RAM), a read-only memory (ROM), an erasable programmable read-only memory (EPROM or Flash memory), a compact disc read-only memory (CD-ROM), and/or some other tangible optical and/or magnetic storage device. In other embodiments of the present disclosure, however, the computer-readable medium may be transitory, such as a propagation signal including computer-executable program code portions embodied therein.

[0213] It will also be understood that one or more computer-executable program code portions for carrying out the specialized operations of the present disclosure may be required on the specialized computer include object-oriented, scripted, and/or unscripted programming languages, such as, for example, Java, Perl, Smalltalk, C++, SAS, SQL, Python, Objective C, and/or the like. In some embodiments, the one or more computer-executable program code portions for carrying out operations of embodiments of the present disclosure are written in conventional procedural programming languages, such as the “C” programming languages and/or similar programming languages. The computer program code may alternatively or additionally be written in one or more multi-paradigm programming languages, such as, for example, F#.

[0214] It will further be understood that some embodiments of the present disclosure are described herein with reference to flowchart illustrations and/or block diagrams of systems, methods, and/or computer program products. It will be understood that each block included in the flowchart illustrations and/or block diagrams, and combinations of blocks included in the flowchart illustrations and/or block diagrams, may be implemented by one or more computer-executable program code portions. These computer-executable program code portions execute via the processor of the computer and/or other programmable data processing apparatus and create mechanisms for implementing the steps and/or functions represented by the flowchart(s) and/or block diagram block(s).

[0215] It will also be understood that the one or more computer-executable program code portions may be stored in a transitory or non-transitory computer-readable medium (e.g., a memory, and the like) that can direct a computer and/or other programmable data processing apparatus to function in a particular manner, such that the computer-executable program code portions stored in the computer-readable medium produce an article of manufacture, including instruction mechanisms which implement the steps and/or functions specified in the flowchart(s) and/or block diagram block(s).

[0216] The one or more computer-executable program code portions may also be loaded onto a computer and/or other programmable data processing apparatus to cause a series of operational steps to be performed on the computer and/or other programmable apparatus. In some embodiments, this produces a computer-implemented process such that the one or more computer-executable program code portions which execute on the computer and/or other programmable apparatus provide operational steps to implement the steps specified in the flowchart(s) and/or the functions specified in the block diagram block(s). Alternatively, computer-implemented steps may be combined with operator and/or human-implemented steps in order to carry out an embodiment of the present disclosure.

[0217] While certain exemplary embodiments have been described and shown in the accompanying drawings, it is to be understood that such embodiments are merely illustrative of, and not restrictive on, the broad disclosure, and that this disclosure not be limited to the specific constructions and

arrangements shown and described, since various other changes, combinations, omissions, modifications, and substitutions, in addition to those set forth in the above paragraphs, are possible. Those skilled in the art will appreciate that various adaptations and modifications of the just described embodiments can be configured without departing from the scope and spirit of the disclosure. Therefore, it is to be understood that, within the scope of the appended claims, the disclosure may be practiced other than as specifically described herein.

INCORPORATION BY REFERENCE

[0218] To supplement the present disclosure, this application further incorporates entirely by reference the following commonly assigned patent applications:

TABLE-US-00001 U.S. patent application Docket Number Ser. No. Title Filed On
14435US1.014033.4624 18/244,832 SYSTEM AND METHOD FOR Sep. 11, 2023
INTERACTIVE AUTOMATED CODE GENERATION AND MODIFICATION FOR DATA
PROCESSING 15179US1.014033.4728 18/244,828 SYSTEM AND METHOD FOR Sep. 11, 2023
DETERMINING DATA FEED SOURCES FOR INTERACTIVE AUTOMATED CODE
GENERATION AND MODIFICATION 15180US1.014033.4729 18/244,776 SYSTEM AND
METHOD FOR Sep. 11, 2023 INTERACTIVE AUTOMATED MODIFICATION OF
TRANSFORMED DATA SETS

Claims

1. A system for determining and maintaining data quality in data processing, the system comprising: at least one processing device at least one non-transitory storage device with computer-readable program code stored thereon, wherein executing the computer-readable code is configured to cause the at least one processing device to perform the following operations: render a flow designer interface including a flow portion and a plugin portion, wherein the flow portion creates a flow operation using blocks connected defining a flow, wherein the blocks are one or more plugins selected from the plugin portion; determine, via the flow designer interface, the one or more plugins used in a flow execution of the flow portion, wherein the flow execution performs at least one transformation on one or more data sets; determine one or more automatic data quality rules based on at least one of the one or more plugins using in the flow execution of the flow designer interface; determine one or more custom data quality rules based on a custom data quality rule input received from an end-point device, wherein the custom data quality rule comprises one or more data statistics to be monitored during the flow execution; cause execution of each of the automatic data quality rules and the one or more custom data quality rules on data in the flow execution; and present, via the flow designer interface, connections between rendering of the one or more plugins used in the flow execution and apply the determined one or more automatic data quality rules based on the connection between the one or more plugins.
2. The system of claim 1, wherein the computer-readable code is further configured to cause the at least one processing device to cause a rendering of a data quality rule interface configured to receive the custom data quality rule input.
3. The system of claim 1, wherein the computer-readable code is further configured to cause the at least one processing device to cause a rendering of results for one or more of the each of the automatic data quality rules and the one or more custom data quality rules on data in the flow execution to the end-point device.
4. The system of claim 1, wherein each of the one or more custom data quality rules comprise a rule category.
5. The system of claim 4, wherein the rule category for each of the one or more custom data quality rules comprises one of accuracy based, completeness based, consistency based, timeliness based, or uniqueness based.
6. The system of claim 1, wherein the one or more automatic data quality rules comprises a column

validation, a cross column validation, and a feed level validation.

7. The system of claim 1, wherein at least one of the one or more automatic data quality rules or the one or more custom data quality rules is configured to determine whether a data set associated with the flow execution contains a required value.

8. A computer program product for determining and maintaining data quality in data processing, the computer program product comprising at least one non-transitory computer-readable medium having computer-readable program code portions embodied therein, the computer-readable program code portions comprising one or more executable portions configured to: render a flow designer interface including a flow portion and a plugin portion, wherein the flow portion creates a flow operation using blocks connected defining a flow, wherein the blocks are one or more plugins selected from the plugin portion; determine, via the flow designer interface, the one or more plugins used in a flow execution of the flow portion, wherein the flow execution performs at least one transformation on one or more data sets; determine one or more automatic data quality rules based on at least one of the one or more plugins using in the flow execution of the flow designer interface; determine one or more custom data quality rules based on a custom data quality rule input received from an end-point device, wherein the custom data quality rule comprises one or more data statistics to be monitored during the flow execution; cause execution of each of the automatic data quality rules and the one or more custom data quality rules on data in the flow execution; and present, via the flow designer interface, connections between rendering of the one or more plugins used in the flow execution and apply the determined one or more automatic data quality rules based on the connection between the one or more plugins.

9. The computer program product of claim 8, wherein the computer-readable program code portions comprising one or more executable portions are also configured to cause a rendering of a data quality rule interface configured to receive the custom data quality rule input.

10. The computer program product of claim 8, wherein the computer-readable program code portions comprising one or more executable portions are also configured to cause a rendering of results for one or more of the each of the automatic data quality rules and the one or more custom data quality rules on data in the flow execution to the end-point device.

11. The computer program product of claim 8, wherein each of the one or more custom data quality rules comprise a rule category.

12. The computer program product of claim 11, wherein the rule category for each of the one or more custom data quality rules comprises one of accuracy based, completeness based, consistency based, timeliness based, or uniqueness based.

13. The computer program product of claim 8, wherein the one or more automatic data quality rules comprises a column validation, a cross column validation, and a feed level validation.

14. The computer program product of claim 8, wherein at least one of the one or more automatic data quality rules or the one or more custom data quality rules is configured to determine whether a data set associated with the flow execution contains a required value.

15. A method for determining and maintaining data quality in data processing, the method comprising: rendering a flow designer interface including a flow portion and a plugin portion, wherein the flow portion creates a flow operation using blocks connected defining a flow, wherein the blocks are one or more plugins selected from the plugin portion; determining, via the flow designer interface, the one or more plugins used in a flow execution of the flow portion, wherein the flow execution performs at least one transformation on one or more data sets; determining one or more automatic data quality rules based on at least one of the one or more plugins using in the flow execution of the flow designer interface; determining one or more custom data quality rules based on a custom data quality rule input received from an end-point device, wherein the custom data quality rule comprises one or more data statistics to be monitored during the flow execution; causing execution of each of the automatic data quality rules and the one or more custom data quality rules on data in the flow execution; and presenting, via the flow designer interface,

connections between rendering of the one or more plugins used in the flow execution and apply the determined one or more automatic data quality rules based on the connection between the one or more plugins.

16. The method of claim 15, further comprising causing a rendering of a data quality rule interface configured to receive the custom data quality rule input.

17. The method of claim 15, further comprising causing a rendering of results for one or more of the each of the automatic data quality rules and the one or more custom data quality rules on data in the flow execution to the end-point device.

18. The method of claim 15, wherein each of the one or more custom data quality rules comprise a rule category, wherein the rule category for each of the one or more custom data quality rules comprises one of accuracy based, completeness based, consistency based, timeliness based, or uniqueness based.

19. The method of claim 15, wherein the one or more automatic data quality rules comprises a column validation, a cross column validation, and a feed level validation.

20. The method of claim 15, wherein at least one of the one or more automatic data quality rules or the one or more custom data quality rules is configured to determine whether a data set associated with the flow execution contains a required value.
